

Structural Principle of Entanglement (PSE)

Foundations of Emergence:
Physical quantities as images of a fundamental pregeometric structure

Advanced Mathematical Formalization



Andrea Toricelli
Independent researcher
Correggio (RE), Italy
a.toricelli@hotmail.it
<https://orcid.org/0009-0001-9024-7888>

Draft dated 03/10/2026
Drafted with the support of artificial intelligence tools
License: CC BY-NC-ND 4.0
Available on Zenodo – **DOI:** 10.5281/zenodo.20218974

Abstract

This work presents a unified mathematical framework—the *Emergent Projection Structure* (PSE)—in which geometry, matter, probability, quantum dynamics, and their mutual unification arise from a single underlying space of emergent states E_{reg} . A global complexity functional $\mathcal{C}(E)$ is constructed on E_{reg} , whose critical values generate the thresholds Θ_k governing the behavior of projection operators Φ . These operators are continuous, compressive, and non-invertible, and their invariants define an emergent geometry $(\mathcal{M}_E, g)$ whose macroscopic limit yields the physical metric g_{phys} . The degeneracy of projection fibers produces informational densities ρ_Φ , scalar fields Ψ , and energy–momentum tensors $T_{\mu\nu}$, recovering Einstein’s equations in the macroscopic regime and Newtonian gravity in the weak-field limit.

Non-invertibility also generates a natural probability measure on fibers, leading to emergent probability distributions, amplitudes ψ , a Hilbert space $\mathcal{H}$, the Born rule, superposition, interference, and a quantum evolution equation whose Schrödinger form appears as a special case. Gravity and quantum theory are shown to be complementary manifestations of the same projection: geometry corresponds to preserved information, while quantum structure corresponds to hidden information. Their relative dominance is governed by the degeneracy ratio $\Delta(E)$, which defines a critical scale ℓ_c marking the transition between geometric, quantum, and unified regimes. Near this scale, the theory predicts corrections to both Einstein’s equations and Schrödinger dynamics.

The PSE thus provides a single emergent architecture in which space-time, matter, probability, and quantum behavior arise from informational properties of E_{reg} . The final principle asserts that observable physics is not fundamental but the result of projecting a deeper emergent structure. The work concludes by outlining the main open problems and a program for experimental verification, including explicit model construction, determination of ℓ_c , and identification of observable deviations from classical and quantum dynamics.

Table of contents

Abstract.....	2
Part I — Foundations of Emerging Complexity	9
Chapter 1 — Foundations of Emerging Complexity.....	9
1.1 Motivation and Objective	9
1.2 Domain of Definition.....	9
1.3 Structural Measure	10
1.4 Metric Complexity	10
1.5 Correlational Complexity	14
1.6 Topological Complexity.....	18
1.7 Global Complexity Function.....	21
1.8 Theory of Critical Thresholds.....	25
Part II — Projection Operators.....	28
Chapter 2 — Mathematical Theory of Projection Operators	28
2.1 Motivation.....	28
2.2 Source and Target Spaces.....	28
2.3 General Definition of Projection Operator	29
2.4 Fundamental Principles.....	29
2.5 Class of Admissible Operators	30
2.6 Projection Entropy	31
2.7 Stability of Operators	31
Chapter 3 — Hierarchy of Projection Operators.....	32
3.1 Projection Complexity	32
3.2 Hierarchy of Projection Classes	32
3.3 Representational Capacity of Operators.....	34
3.4 Projection Levels	35
Chapter 4 — Constructive Construction of Projection Operators	36
4.1 Motivation.....	36
4.2 Structural Compression Operators.....	37
4.3 Linear Class	37
4.4 Non-Linear Class	38
4.5 Non-Invertibility Theorem	38
4.6 Fundamental Operator of the PSE	39
Part III — Emerging Geometry and Dynamics.....	40
Chapter 5 — Emerging Geometry of Projection	40
5.1 Motivation.....	40

5.2 Emerging Coordinates	40
5.3 Induced Emergent Distance	41
5.4 Emerging Information Tensor	41
5.5 Principle of Geometric Emergence	42
Chapter 6 — Dynamics of Emerging Complexity	42
6.1 Evolutionary Families	42
6.2 Complexity Velocity	42
6.3 Principle of Minimal Evolution	43
6.4 Stability of Thresholds	44
Part IV – Mathematical Consolidation	45
C1 – Completeness of the Emergent Space	45
C2 – Regularity of the Complexity Functionals	47
C2.1 Regularity of Metric Complexity	48
C2.3 Stability of Quantitative Components	49
C2.4 Consequences for the Theory	49
C3 – Topological Stability and Regularity of Topological Complexity	50
C3.1 Sensitivity of Emerging Topology	50
C3.2 Homological Stability	51
C3.3 Regularity of Topological Complexity	52
C3.4 Critical Regimes.....	53
C4 – Rigorous Foundations of the Theory of Projection Operators	53
C4.1 Regularity of Projection Operators	54
C4.2 Stability of Projections	55
C4.3 Structure of Projection Fibers	55
C4.4 Compatibility with Complexity Classes	56
C5 – Rigorous Foundations of Emerging Geometry	57
C5.1 Differentiable Structure of the Emerging Manifold	57
C5.2 Emerging Metric	58
C5.3 Emerging Geometric Distance	59
C5.5 Geometric Correspondence Principle	60
C6 – Rigorous Foundations of the Dynamics of Complexity	61
C6.1 Regularity of Evolutionary Trajectories	61
C6.2 Complexity Action.....	62
C6.3 Derivation of the Dynamical Equations	63
C6.4 Conservation of Complexity Energy	63
C6.5 Stability of Critical Thresholds	64

C6.6 Transition Regimes.....	64
C7 – Global Coherence of the Emerging Chain.....	65
C7.1 Fundamental Emerging Chain	65
C7.2 Functional Compatibility.....	65
C7.3 Structural Coherence.....	66
C7.4 Emerging Closure.....	66
C7.5 Principle of Mathematical Self-Sufficiency.....	66
Part V – Emergence of the Physical Metric	68
Chapter 7 – Emergence of the Physical Metric	68
7.1 Geometric Observability Condition	68
7.2 Observable Distance	68
7.3 Macroscopic Geometric Limit	69
7.4 Emerging Physical Metric	70
7.5 Theorem 7.2 – Existence of a Limit Geometry.....	70
7.6 Principle of Emergence of Space-Time	70
Chapter 8 – Dynamics of the Emerging Physical Metric.....	71
8.1 Motivation.....	71
8.2 Emerging Physical Manifold	71
8.3 Emerging Curvature.....	71
8.4 Emerging Geometric Action	72
8.5 Minimal Lagrangian	72
8.6 Theorem 8.1 – Emerging Geometric Equations.....	73
8.7 Emerging Interpretation	73
Chapter 9 – Emergence of Matter and of the Energy–Momentum Tensor	73
9.1 Motivation.....	73
9.2 Informational Loss and Projection Residue.....	73
Definition 9.1 – Emerging Degeneracy Density	74
9.3 Emerging Geometric Entropy	74
9.4 Emerging Matter Field	74
9.5 Emerging Energy–Momentum Tensor.....	74
9.6 Geometric Equations with Source	75
Part VI – Physical Limits and Correspondence Principle	76
Chapter 10 – Relativistic Limit of the PSE.....	76
10.1 Motivation.....	76
10.2 Geometric Reduction	76
10.3 Reduction of the Emerging Equations	77

10.4 Relativistic Correspondence Principle	77
10.5 Interpretation of the Gravitational Field	77
Chapter 11 – Physical Properties of Emerging Matter	78
11.1 Motivation	78
11.2 Emerging Energy Density	78
11.3 Emerging Pressure	78
11.4 Emerging Equation of State	79
11.5 Local Conservation	79
11.6 Energy Conditions	80
11.7 Emerging Mass	80
11.8 Physical Interpretation	80
Chapter 12 – Newtonian Limit of the PSE	81
12.1 Motivation	81
12.2 Weak-Field Regime	81
12.3 Emerging Gravitational Potential	81
12.4 Static Limit	82
12.5 Emerging Poisson Equation	82
12.6 Emerging Gravitational Mass	82
12.7 Newtonian Correspondence Principle	83
Part VII – Emergence of Probability and Quantum Dynamics	85
Chapter 13 – Emergence of Probability	85
13.1 Motivation	85
13.2 Measure on Fibers	85
13.3 Emerging Probability	85
13.4 Projection Entropy and Information	86
13.5 Emerging Uncertainty Principle	86
Chapter 14 – Emergence of the Hilbert Space and Probability Amplitudes	86
14.1 Motivation	86
14.2 Equivalent Emerging States	87
14.3 Square Root of Probability	87
14.4 Emerging State Space	88
14.5 Emerging Superposition Principle	88
14.6 Emerging Interference	89
14.7 Emerging Born Rule	89
Chapter 15 – Emerging Quantum Dynamics	90
15.1 Motivation	90

15.2 Evolution of Emerging Fibers	90
15.3 Conservation of Probability	90
15.4 Evolution Generator Operator	91
15.5 Emerging Quantum Equation	91
15.6 Schrödinger Limit	92
15.7 Interpretation of the Superposition Principle.....	92
15.8 Complete Quantum Chain	93
Part VIII – Emerging Unification of Gravity and Quantum Theory	94
Chapter 16 – Principle of Emerging Unification.....	94
16.1 Motivation	94
16.2 Dual Manifestation of the Projection.....	94
16.3 Principle of Emerging Duality.....	95
16.4 Total Amount of Information.....	95
16.5 Common Origin of Mass and Probability	96
16.6 Principle of Emerging Unification.....	96
Chapter 17 – Critical Regimes and Transitions Between Geometry and Quantum Theory.....	97
17.1 Motivation	97
17.2 Degeneracy Ratio	97
17.3 Geometric Regime.....	97
17.4 Quantum Regime	98
17.5 Critical Regime.....	98
17.6 Emerging Critical Length	99
17.7 Diagram of Emerging Regimes.....	99
Chapter 18 – Origin of the Emerging Critical Scale.....	99
18.1 Motivation	99
18.2 Geometric Complexity and Quantum Complexity	100
18.3 Informational Equilibrium Point.....	100
18.4 Origin of the Critical Scale	101
18.5 Stability of the Critical Scale	101
18.6 Interpretation of the Planck Scale.....	102
18.7 Unification Region	102
Chapter 19 – Dynamics of Unification and Predictions of the PSE	103
19.1 Motivation	103
19.2 Unification Parameter	103
19.3 Dynamics of Unification.....	103
19.4 Stability of the Critical Region	104

19.5 Emerging Corrections to Gravity	104
19.6 Emerging Corrections to Quantum Dynamics	105
19.7 First Qualitative Predictions of the PSE	105
19.8 Diagram of the Complete Architecture	105
Chapter 20 – General Conclusions, Limits of the Theory, and Verification Program	106
20.1 Motivation	106
20.2 Fundamental Results.....	106
20.3 Final Architecture of the Theory	108
20.4 Current Limits of the Theory	108
20.5 Program for Physical Verification	109
20.6 Final Principle of the PSE	109
20.7 General Conclusion.....	109

Part I — Foundations of Emerging Complexity

Chapter 1 — Foundations of Emerging Complexity

1.1 Motivation and Objective

In the Axiomatic Mathematical Formalization of the PSE, the appearance of the projection classes $\Phi_1, \Phi_2, \dots, \Phi_6$ is associated with a succession of structural thresholds $\theta_1 < \theta_2 < \dots < \theta_6$.

These thresholds are introduced as primitive objects of the theory and represent the levels of complexity required for a given projection class to be definable.

From an axiomatic standpoint, this formulation is sufficient, as it guarantees the logical coherence of the ontological hierarchy of the PSE.

From a constructive standpoint, however, a fundamental question remains open: **how is the complexity of an emergent structure measured?**

The present section introduces a quantitative theory of emerging complexity with the purpose of:

- defining a mathematical measure of the complexity of the regularized emergent domain;
- constructing a functional space in which such complexity is well defined;
- proving the existence, finiteness, and stability of the complexity measure;
- preparing the rigorous derivation of the thresholds θ_k as critical values of a structural function.

The final objective is to transform the complexity thresholds from axiomatic objects into mathematically determined quantities.

1.2 Domain of Definition

Emerging complexity is defined on the regularized emergent domain

$$E_{\text{reg}} = \left(X, D_{S,\text{reg}}^{(e)}, C_{e,\text{reg}} \right)$$

where:

- X is the continuous emergent domain;
- $D_{S,\text{reg}}^{(e)}: X \times X \rightarrow \mathbb{R}^+$ is the regularized emergent distance;
- $C_{e,\text{reg}}: X \times X \rightarrow \mathbb{R}$ is the regularized emergent correlation.

Hypothesis H1 — Compactness

The topological space X is assumed to be compact.

Hypothesis H2 — Regularity

The functions $D_{S,\text{reg}}^{(e)}$ and $C_{e,\text{reg}}$ are continuous on $X \times X$.

Hypothesis H3 — Boundedness

There exist finite constants $D_{\text{max}} > 0$ and $C_{\text{max}} > 0$ such that

$$D_{S,\text{reg}}^{(e)}(x, y) \leq D_{\max}, \quad |C_{e,\text{reg}}(x, y)| \leq C_{\max}$$

for every $(x, y) \in X \times X$.

Observation

The above hypotheses are already compatible with the Axiomatic Formalization of the PSE. Therefore, no new ontological assumption is introduced.

This theory simply employs, in quantitative form, properties already present in the regularized emergent structure.

1.3 Structural Measure

Since the emergent domain is compact, there exists at least one finite Borel measure μ defined on X .

In what follows, we use a normalized measure:

$$\mu(X) = 1.$$

This normalization allows all subsequent integrals to be interpreted as global structural averages.

Definition 1.1 — Regularized Structural Space

The regularized structural space is defined as

$$E_{\text{reg}} = C^0(X \times X) \times C^0(X \times X)$$

equipped with the norm

$$\|(D, C)\|_{E_{\text{reg}}} = \max(\|D\|_{\infty}, \|C\|_{\infty}).$$

The elements of E_{reg} represent all possible regularized emergent structures compatible with the PSE.

Proposition 1.1

The space E_{reg} is a Banach space.

Proof

The space $C^0(X \times X)$ is complete with respect to the uniform norm because $X \times X$ is compact.

The product of two Banach spaces is again a Banach space.

Therefore,

$$E_{\text{reg}} = C^0(X \times X) \times C^0(X \times X)$$

is complete.

1.4 Metric Complexity

Introduction

The first component of emerging complexity measures the degree of articulation of the structure encoded by the regularized emergent distance

$$D_{S,\text{reg}}^{(e)}.$$

Since in the PSE every subsequent manifestation of physical structure depends on the emergent geometry contained in the regularized distance, it is natural to assume that part of the total complexity is associated with the global distribution of the values of

$$D_{S,\text{reg}}^{(e)}.$$

The objective of this section is to construct a quantitative measure that:

- is well defined on the compact emergent domain;
- is finite;
- is stable under small perturbations;
- is invariant under fundamental isomorphisms;
- is compatible with the axiomatic structure of the PSE.

Definition 1.2 — Metric Structural Weight

A metric structural weight is defined as a function

$$w_D: X \times X \rightarrow \mathbb{R}^+$$

such that:

$$w_D(x, y) \geq 0$$

for every $(x, y) \in X \times X$, and

$$\int_{X \times X} w_D(x, y) d\mu(x) d\mu(y) = 1.$$

The weight represents the relative importance of different sectors of the emergent structure.

Canonical Case

In the simplest case one assumes:

$$w_D(x, y) = 1.$$

Since $\mu(X) = 1$, it follows immediately that:

$$\int_{X \times X} w_D(x, y) d\mu(x) d\mu(y) = 1.$$

This case describes a completely isotropic structure.

All subsequent results remain valid for more general weights.

Definition 1.3 — Metric Complexity

The metric complexity of the regularized emergent domain

$$E_{\text{reg}} = \left(X, D_{S,\text{reg}}^{(e)}, C_{e,\text{reg}} \right)$$

is defined as the functional

$$C_D(E_{\text{reg}}) = \int_{X \times X} w_D(x, y) D_{S,\text{reg}}^{(e)}(x, y) d\mu(x) d\mu(y).$$

The quantity C_D represents the average structural content associated with the regularized emergent distance.

Interpretation

If the regularized distance takes almost constant values over the entire domain, the metric complexity is reduced.

Conversely, a rich and articulated distribution of the emergent distance yields higher values of C_D .

Metric complexity therefore measures the global richness of the emergent organization encoded in

$$D_{S,\text{reg}}^{(e)}.$$

Theorem 1.1 — Existence of Metric Complexity

Under hypotheses H1–H3, the quantity

$$C_D(E_{\text{reg}})$$

is well defined.

Proof

By hypothesis,

$$D_{S,\text{reg}}^{(e)} \in C^0(X \times X).$$

Since X is compact, continuity implies boundedness of the function. Therefore, there exists a constant $D_{\max}$ such that

$$0 \leq D_{S,\text{reg}}^{(e)}(x, y) \leq D_{\max}.$$

The structural weight w_D is non-negative and integrable.

It follows that the integrand

$$w_D(x, y) D_{S,\text{reg}}^{(e)}(x, y)$$

is measurable and bounded.

The Lebesgue integral is therefore finite:

$$0 \leq C_D(E_{\text{reg}}) < \infty.$$

Hence, metric complexity is well defined.

Corollary 1.1 — Boundedness of Metric Complexity

The estimate

$$0 \leq C_D(E_{\text{reg}}) \leq D_{\max}$$

holds.

Proof

From the definition:

$$C_D = \int_{X \times X} w_D(x, y) D_{S, \text{reg}}^{(e)}(x, y) d\mu(x) d\mu(y).$$

Since

$$D_{S, \text{reg}}^{(e)} \leq D_{\max},$$

it follows that

$$C_D \leq D_{\max} \int_{X \times X} w_D(x, y) d\mu(x) d\mu(y).$$

Using the normalization of the weight:

$$\int_{X \times X} w_D d\mu d\mu = 1,$$

we obtain:

$$C_D \leq D_{\max}.$$

Theorem 1.2 — Invariance under Fundamental Isomorphisms

Let

$$\sigma: X \rightarrow X$$

be an isomorphism of the emergent domain.

Then:

$$C_D(E_{\text{reg}}^\sigma) = C_D(E_{\text{reg}}).$$

Proof

By definition of the action of an isomorphism:

$$D_{S, \text{reg}}^{(e)\sigma}(x, y) = D_{S, \text{reg}}^{(e)}(\sigma(x), \sigma(y)).$$

Since isomorphisms preserve the emergent structure and the normalized measure, the change of variables in the integral leaves the value of the functional invariant.

Thus:

$$\int_{X \times X} w_D(x, y) D_{S, \text{reg}}^{(e)\sigma}(x, y) d\mu(x) d\mu(y) = \int_{X \times X} w_D(x, y) D_{S, \text{reg}}^{(e)}(x, y) d\mu(x) d\mu(y).$$

Hence:

$$C_D(E_{\text{reg}}^\sigma) = C_D(E_{\text{reg}}).$$

Ontological Meaning

Metric complexity does **not** represent a physical quantity.

It is an internal property of the regularized emergent structure.

Consequently, C_D does **not** measure:

- physical distances;

- energy;
- thermodynamic entropy;
- observable information.

It measures exclusively the degree of structural organization contained in the regularized emergent distance.

1.5 Correlational Complexity

Introduction

The regularized emergent distance describes the geometry of the emergent structure. However, in the PSE, geometry alone does not exhaust the organizational content of the emergent domain.

The regularized emergent correlation

$$C_{e,\text{reg}}$$

encodes structural connections that cannot be reduced to mere metric separation between elements of the domain.

It follows that a quantitative theory of complexity cannot depend exclusively on

$$D_{S,\text{reg}}^{(e)},$$

but must also incorporate the contribution associated with emergent correlations.

The purpose of this section is to define this contribution and to prove its fundamental properties.

1.5.1 Emergent Correlation and Structural Organization

Let

$$E_{\text{reg}} = (X, D_{S,\text{reg}}^{(e)}, C_{e,\text{reg}})$$

be a regularized emergent domain.

The emergent correlation

$$C_{e,\text{reg}}(x, y)$$

measures the intensity of the structural connection between two elements $x, y \in X$.

In general:

$$C_{e,\text{reg}}: X \times X \rightarrow \mathbb{R}.$$

Since both positive and negative correlations represent structural information, the complexity measure must depend on the absolute value of the correlation.

Definition 1.4 — Absolute Correlational Density

The absolute correlational density is defined as the function

$$\rho_C(x, y) = |C_{e,\text{reg}}(x, y)|.$$

It represents the structural intensity of the correlation independently of its sign.

Observation

From the ontological standpoint of the PSE, the sign of the correlation may distinguish different modes of organization. However, in the complexity measure, what matters is the total amount of correlational structure available.

For this reason, the definition uses the absolute value.

1.5.2 Definition of Correlational Complexity

Definition 1.5 — Correlational Complexity

The correlational complexity of the regularized emergent domain is defined as the functional

$$C_C(E_{\text{reg}}) = \int_{X \times X} |C_{e,\text{reg}}(x, y)| d\mu(x) d\mu(y).$$

Equivalently:

$$C_C(E_{\text{reg}}) = \int_{X \times X} \rho_C(x, y) d\mu(x) d\mu(y).$$

Interpretation

The quantity C_C measures the global average amount of correlation present within the emergent structure.

- Small values of C_C correspond to weakly organized emergent domains.
- Large values of C_C indicate a rich network of structural connections.

Theorem 1.3 — Existence of Correlational Complexity

Under hypotheses H1–H3, the quantity

$$C_C(E_{\text{reg}})$$

is well defined.

Proof

By hypothesis H2,

$$C_{e,\text{reg}}$$

is continuous on $X \times X$.

The continuity of the absolute value implies that

$$|C_{e,\text{reg}}|$$

is also continuous.

Since X is compact, the function

$$|C_{e,\text{reg}}|$$

is bounded.

Thus, there exists a constant $C_{\text{max}} > 0$ such that

$$|C_{e,\text{reg}}(x, y)| \leq C_{\max}$$

for every $(x, y) \in X \times X$.

The integrand is therefore bounded and measurable.

It follows that the Lebesgue integral exists and is finite.

Hence, correlational complexity is well defined.

Corollary 1.2 — Positivity

For every regularized emergent domain,

$$C_C(E_{\text{reg}}) \geq 0.$$

Proof

The integrand

$$|C_{e,\text{reg}}|$$

is non-negative.

The integral of a non-negative function is non-negative.

Corollary 1.3 — Boundedness

The estimate

$$0 \leq C_C(E_{\text{reg}}) \leq C_{\max}$$

holds.

Proof

Since

$$|C_{e,\text{reg}}(x, y)| \leq C_{\max},$$

we have

$$C_C = \int |C_{e,\text{reg}}| d\mu d\mu \leq C_{\max} \int d\mu d\mu.$$

Since the measure is normalized,

$$\mu(X) = 1,$$

it follows that

$$C_C \leq C_{\max}.$$

Theorem 1.4 — Continuity of the Correlational Functional

The functional

$$C_C: E_{\text{reg}} \rightarrow \mathbb{R}$$

is continuous with respect to the uniform norm on E_{reg} .

Proof

Let $C_n \rightarrow C$ in the uniform norm.

Then:

$$\|C_n - C\|_\infty \rightarrow 0.$$

Since:

$$\| |C_n| - |C| \| \leq \|C_n - C\|,$$

we obtain:

$$\| |C_n| - |C| \| \leq \|C_n - C\|_\infty.$$

Integrating:

$$|C_C(C_n) - C_C(C)| \leq \|C_n - C\|_\infty.$$

The right-hand side tends to zero.

Thus:

$$C_C(C_n) \rightarrow C_C(C).$$

Theorem 1.5 — Invariance under Structural Isomorphisms

Let

$$\sigma: X \rightarrow X$$

be an isomorphism of the emergent structure.

Then:

$$C_C(E_{\text{reg}}^\sigma) = C_C(E_{\text{reg}}).$$

Proof

By definition:

$$C_{e,\text{reg}}^\sigma(x, y) = C_{e,\text{reg}}(\sigma(x), \sigma(y)).$$

The isomorphism preserves both the measure and the structure of the domain.

Applying the change-of-variables theorem:

$$\int |C_{e,\text{reg}}^\sigma| d\mu d\mu = \int |C_{e,\text{reg}}| d\mu d\mu.$$

Therefore:

$$C_C(E_{\text{reg}}^\sigma) = C_C(E_{\text{reg}}).$$

Ontological Meaning of Correlational Complexity

Correlational complexity quantifies the total amount of interdependence present in the emergent structure.

It does **not** represent:

- observable quantum entanglement;

- measurable physical information;
- experimental statistical correlation.

It represents exclusively a **pre-physical property** of emergent organization.

Within the PSE framework, correlational complexity constitutes the **second pillar** of the forthcoming global complexity function.

The next section will introduce the third fundamental contribution: **topological complexity**, which will allow the global structure of emergent connections to be measured through topological invariants.

1.6 Topological Complexity

Introduction

The components introduced in the previous sections measure local or average aspects of the emergent structure. Metric complexity quantifies the organization associated with the regularized emergent distance. Correlational complexity quantifies the global intensity of emergent correlations.

However, two emergent domains may possess similar values of metric and correlational complexity while exhibiting profoundly different structural configurations.

It is therefore necessary to introduce a measure capable of describing the global shape of the network of emergent connections. This information is provided by topology.

Topological complexity thus represents the third fundamental contribution to the quantitative theory of emerging complexity.

1.6.1 Associated Emergent Graph

The correlational information contained in

$$C_{e,\text{reg}}$$

can be transformed into a combinatorial structure.

Definition 1.6 — Correlational Activation Level

Let

$$\eta > 0$$

be a fixed correlational threshold.

Define the adjacency relation:

$$x \sim_{\eta} y \text{ if and only if } |C_{e,\text{reg}}(x, y)| \geq \eta.$$

Definition 1.7 — Emergent Graph

The emergent graph associated with the threshold η is defined as

$$G_{\eta}(E_{\text{reg}}) = (V, E)$$

where:

- $V = X$,
- $E = \{(x, y) \in X \times X : |C_{e,\text{reg}}(x, y)| \geq \eta\}.$

Each edge represents a structural connection sufficiently strong to survive the activation threshold.

Interpretation

The threshold η acts as a structural filter.

- Large values select only the strongest correlations.
- Smaller values allow one to observe a richer and more detailed network.

The emergent topology therefore depends on the chosen observation scale.

1.6.2 Associated Simplicial Complex

To study the global structure of the graph, it is not sufficient to consider only the edges. One must introduce objects capable of describing higher-order connections.

For this purpose, we use the associated simplicial complex.

Definition 1.8 — Flag Complex

To the graph

$$G_\eta(E_{\text{reg}})$$

one associates the simplicial complex

$$K_\eta(E_{\text{reg}})$$

constructed by assigning:

- a vertex to each node;
- a segment to each edge;
- a triangle to each fully connected triple;
- an n -simplex to each clique of cardinality $n + 1$.

This construction is known as the **flag complex**.

Observation

The structure of the complex contains global information not visible by observing only the graph. It allows one to detect:

- connected components,
- cycles,
- cavities,
- multidimensional structures.

1.6.3 Topological Invariants

Topological complexity will be constructed using the fundamental invariants of algebraic topology.

Definition 1.9 — Betti Numbers

For each integer $k \geq 0$, the k -th Betti number is defined as

$$\beta_k = \dim H_k(K_\eta),$$

where $H_k(K_\eta)$ is the homology group of order k .

Geometric Interpretation

The first Betti numbers have a clear interpretation:

- β_0 measures the number of connected components.
- β_1 measures the number of independent cycles.
- β_2 measures the number of two-dimensional cavities.
- β_3 measures the three-dimensional cavities.

and so on.

1.6.4 Definition of Topological Complexity

The previous quantities allow one to define a comprehensive measure of topological richness.

Definition 1.10 — Topological Complexity

The topological complexity of the emergent domain is defined as

$$C_T(E_{\text{reg}}) = \sum_{k=0}^{\infty} \omega_k \beta_k$$

where:

- $\omega_k > 0$ are weight coefficients such that

$$\sum_{k=0}^{\infty} \omega_k < \infty.$$

Canonical Case

In the simplest case one chooses:

$$\omega_k = 2^{-k}.$$

This yields:

$$C_T = \beta_0 + \frac{1}{2}\beta_1 + \frac{1}{4}\beta_2 + \frac{1}{8}\beta_3 + \dots$$

which guarantees convergence of the series.

Theorem 1.6 — Existence of Topological Complexity

Under hypotheses H1–H3, the quantity

$$C_T(E_{\text{reg}})$$

is well defined.

Proof

Since the domain is compact and

$$C_{e,\text{reg}}$$

is continuous, the emergent graph associated with a fixed threshold possesses a well-defined simplicial structure.

The homology groups are therefore finitely generated.

It follows that all Betti numbers are finite.

Since

$$\sum_{k=0}^{\infty} \omega_k$$

converges and each term is non-negative, the series defining C_T converges absolutely.

Thus:

$$0 \leq C_T(E_{\text{reg}}) < \infty.$$

Theorem 1.7 — Topological Invariance

Topological complexity is invariant under structural isomorphisms.

Proof

Isomorphisms of the emergent structure induce isomorphisms between the associated simplicial complexes.

Homology is a topological invariant.

Consequently, all Betti numbers remain unchanged.

Since C_T depends exclusively on Betti numbers, it follows that:

$$C_T(E_{\text{reg}}^\sigma) = C_T(E_{\text{reg}}).$$

Ontological Meaning

Topological complexity measures the global richness of the emergent structure.

Unlike the previous components, it does not depend on the average intensity of correlations or distances, but on the overall shape of the organizational network.

Within the PSE framework:

- metric complexity measures the structure of separations;
- correlational complexity measures the structure of connections;
- topological complexity measures the structure of global configurations.

The three components constitute the necessary building blocks for constructing the emergent complexity function.

1.7 Global Complexity Function

Introduction

The previous sections introduced three distinct contributions to the quantitative description of the emergent structure:

- metric complexity C_D ;
- correlational complexity C_C ;
- topological complexity C_T .

Each of these quantities captures a different aspect of emergent organization. However, the transition mechanism between the different ontological levels of the PSE cannot depend on a single component.

It is therefore necessary to introduce a global measure capable of integrating these contributions into a single structural functional. This quantity will constitute the central object of the quantitative theory of emerging complexity.

The future thresholds Θ_k will be defined in terms of the properties of this functional.

1.7.1 Definition of Total Complexity

Definition 1.11 — Structural Coefficients

Introduce coefficients

$$\alpha_D, \alpha_C, \alpha_T \in \mathbb{R}^+$$

such that

$$\alpha_D + \alpha_C + \alpha_T = 1.$$

They represent the relative weight assigned to the three fundamental components of complexity.

Observation

The coefficients have no physical meaning. They represent purely structural parameters of the theory. Different values allow one to emphasize different aspects of emergent organization.

Definition 1.12 — Global Complexity Function

The global complexity function is defined as the functional

$$C: E_{\text{reg}} \rightarrow \mathbb{R}$$

given by

$$C(E_{\text{reg}}) = \alpha_D C_D(E_{\text{reg}}) + \alpha_C C_C(E_{\text{reg}}) + \alpha_T C_T(E_{\text{reg}}).$$

Equivalently:

$$C = \alpha_D C_D + \alpha_C C_C + \alpha_T C_T.$$

Interpretation

The quantity

$$C(E_{\text{reg}})$$

measures the total amount of organization present in the regularized emergent domain.

It constitutes the first global quantity of the PSE capable of simultaneously incorporating:

- structural separations,
- structural connections,
- topological organization.

1.7.2 Fundamental Properties

Theorem 1.8 — Existence of Total Complexity

For every regularized emergent domain E_{reg} , the quantity

$$C(E_{\text{reg}})$$

is well defined.

Proof

From Theorems 1.1, 1.3, and 1.6 it follows that:

$$C_D < \infty, C_C < \infty, C_T < \infty.$$

Since $\alpha_D, \alpha_C, \alpha_T$ are finite constants, the linear combination

$$C = \alpha_D C_D + \alpha_C C_C + \alpha_T C_T$$

is also finite.

Therefore:

$$0 \leq C(E_{\text{reg}}) < \infty.$$

Theorem 1.9 — Positivity

For every regularized emergent domain,

$$C(E_{\text{reg}}) \geq 0.$$

Proof

The quantities C_D, C_C, C_T are non-negative. Moreover, $\alpha_D, \alpha_C, \alpha_T$ are positive.

A linear combination of non-negative quantities with positive coefficients is non-negative.

Corollary 1.4

The global complexity function is bounded below. In particular:

$$\inf C = 0.$$

1.7.3 Continuity of Complexity

The stability of the theory requires that small variations in the emergent structure do not produce arbitrary jumps in complexity.

Theorem 1.10 — Continuity of the Global Functional

The functional

$$C: E_{\text{reg}} \rightarrow \mathbb{R}$$

is continuous with respect to the norm on E_{reg} .

Proof

From the previous theorems, C_D and C_C are continuous. Assuming a fixed topological threshold, C_T is also stable under sufficiently small perturbations that do not alter the homology class of the associated complex.

A linear combination of continuous functionals is continuous.

Therefore, C is continuous.

Meaning

This result guarantees that complexity is a structurally robust quantity. Small deformations of the emergent state produce only small variations in complexity. The theory is thus immune to pathological instabilities.

1.7.4 Structural Invariance

A fundamental requirement of the PSE is that quantitative objects depend on emergent organization and not on the representation used.

Theorem 1.11 — Invariance under Isomorphisms

Let

$$\sigma: X \rightarrow X$$

be an isomorphism of the emergent structure.

Then:

$$C(E_{\text{reg}}^\sigma) = C(E_{\text{reg}}).$$

Proof

From Theorems 1.2, 1.5, and 1.7:

$$C_D(E^\sigma) = C_D(E), C_C(E^\sigma) = C_C(E), C_T(E^\sigma) = C_T(E).$$

Therefore:

$$C(E^\sigma) = \alpha_D C_D(E) + \alpha_C C_C(E) + \alpha_T C_T(E) = C(E).$$

*1.7.5 Class of Zero Complexity***Definition 1.13 — Structurally Trivial State**

An emergent domain is said to be structurally trivial if

$$D_{S,\text{reg}}^{(e)}(x, y) = 0 \text{ and } C_{e,\text{reg}}(x, y) = 0$$

for every $x, y \in X$.

Theorem 1.12

For a structurally trivial state,

$$C(E_{\text{reg}}) = 0.$$

Proof

In this situation:

$$C_D = 0, C_C = 0.$$

The emergent graph contains no active connections, and the associated complex has zero topological complexity. Thus:

$$C_T = 0.$$

It follows that:

$$C = \alpha_D \cdot 0 + \alpha_C \cdot 0 + \alpha_T \cdot 0 = 0.$$

1.8 Theory of Critical Thresholds

1.8.1 Motivation

In the Axiomatic Formalization of the PSE, the hierarchy of projection classes

$$\Phi_1, \Phi_2, \dots, \Phi_6$$

is associated with an increasing sequence of structural thresholds

$$\theta_1 < \theta_2 < \dots < \theta_6.$$

These thresholds delimit different regimes of organization of the emergent domain.

The objective of this section is to show how such thresholds can be obtained as critical values of the global complexity function

$$C(E_{\text{reg}}).$$

1.8.2 Family of Emergent States

Definition 1.14

Consider a parameterized family of emergent domains

$$E(\lambda) = (X, D_\lambda, C_\lambda)$$

with $\lambda \in I$ and $I \subset \mathbb{R}$ a connected interval.

Assume that:

$$D_\lambda \text{ and } C_\lambda$$

depend continuously on λ .

Definition 1.15

The complexity function associated with the family is

$$\mathcal{C}(\lambda) = C(E(\lambda)).$$

It describes the evolution of structural complexity as the generative parameter varies.

Observation

In the context of the PSE, the parameter λ does not necessarily represent time. It may represent any process of structural accumulation capable of modifying the emergent domain.

1.8.3 Structural Transitions

The fundamental idea is that not every increase in complexity produces a new projection class. The appearance of a new class requires a qualitative modification of the emergent structure.

Definition 1.16

A structural transition occurs at the point λ_c if at least one of the following quantities undergoes a qualitative change:

- the topology of K_η ;
- the stability class of correlations;
- the set of admissible projections.

Interpretation

A structural transition is not defined by a mere quantitative increase in complexity. It corresponds to a change of organizational regime.

In this sense, the threshold theory of the PSE is analogous to the mathematical theory of bifurcations.

1.8.4 Critical Points of Complexity

Definition 1.17

A value λ_c is called *critical* if:

$$\frac{dC}{d\lambda}$$

is non-regular, or if a structural property changes at λ_c .

Definition 1.18

A *critical complexity value* is defined as:

$$\Theta = C(\lambda_c).$$

Observation

The quantity Θ is not postulated. It emerges from the dynamics of the structure.

The threshold is therefore an *effect* of the theory, not a primitive ingredient.

Theorem 1.13 — Existence of Critical Values

Let $C(\lambda)$ be continuous on a compact interval $[a, b]$. If the family $E(\lambda)$ exhibits a finite number of structural transitions

$$\lambda_1, \lambda_2, \dots, \lambda_n,$$

then there exists a finite sequence of critical values

$$\Theta_1, \Theta_2, \dots, \Theta_n$$

defined by

$$\Theta_k = C(\lambda_k).$$

Proof

By definition of structural transition, the points λ_k are distinct. The continuity of C guarantees that $C(\lambda_k)$ is well defined for each k .

Therefore, each transition induces a critical complexity value.

1.8.5 Hierarchy of Thresholds

Definition 1.19

The *structural hierarchy* is the ordered sequence

$$\theta_1 < \theta_2 < \dots < \theta_n.$$

Each interval

$$[\theta_k, \theta_{k+1})$$

identifies a stable structural regime.

Theorem 1.14 — Regime Stability

If no structural transitions occur in the interval

$$(\theta_k, \theta_{k+1}),$$

then the class of admissible projections remains invariant.

Proof

By definition, a variation in the class of projections constitutes a structural transition. The absence of transitions therefore implies invariance of the admissible class.

Conclusion of the Section

The theory now possesses a conceptually powerful result:

The thresholds θ_k are no longer introduced as primitive constants of the theory, but as images of the critical points of the global complexity function.

In the language of the advanced mathematics of the PSE:

$$\boxed{\theta_k = C(E(\lambda_k))}$$

where λ_k is the point at which a structural transition of the emergent domain occurs.

Part II — Projection Operators

Chapter 2 — Mathematical Theory of Projection Operators

2.1 Motivation

The complexity theory developed in Chapter 1 allowed us to introduce a hierarchy of critical thresholds

$$\theta_1 < \theta_2 < \dots < \theta_n.$$

It now remains to rigorously construct the central object that connects the emergent domain to the physically observable domain. In the axiomatic formulation, this role is played by the projection operators

$$\Phi.$$

They represent the mechanism through which an emergent structure is transformed into a physically accessible representation.

The purpose of this chapter is to develop an explicit mathematical theory of projection operators, proving their:

- existence;
- structural properties;
- stability;
- dependence on complexity;
- classification into hierarchical classes.

2.2 Source and Target Spaces

Definition 2.1 — Emergent Space

Let

$$E_{\text{reg}}$$

be the space of regularized emergent structures introduced in Chapter 1:

$$E_{\text{reg}} = C^0(X \times X) \times C^0(X \times X).$$

A generic element will be denoted by

$$E = (D, C).$$

Definition 2.2 — Space of Projected Configurations

Define a space of physical configurations

$$\mathcal{P}.$$

The elements of $\mathcal{P}$ represent structures accessible after projection. In what follows, we assume that $\mathcal{P}$ is a complete metric space.

Observation

The theory does not initially require $\mathcal{P}$ to be an ordinary physical space. Spacetime may eventually emerge as a particular structure belonging to $\mathcal{P}$.

2.3 General Definition of Projection Operator

Definition 2.3

A projection operator is a map

$$\Phi: E_{\text{reg}} \rightarrow \mathcal{P}.$$

Interpretation

The operator

$$E \mapsto \Phi(E)$$

associates an emergent structure with an observable configuration.

The projection does not represent a physical transformation. It represents a structural reduction.

2.4 Fundamental Principles

Projection operators admitted by the PSE must satisfy precise properties. These properties replace the informal axioms present in the original formulation.

Axiom P1 — Monotonicity

If

$$\mathcal{C}(E_1) \leq \mathcal{C}(E_2),$$

then

$$\mathcal{I}(\Phi(E_1)) \leq \mathcal{I}(\Phi(E_2)),$$

where $\mathcal{I}$ measures the structural content of the projected configuration.

Meaning

A more complex emergent structure cannot generate a structurally poorer image.

Axiom P2 — Continuity

For every sequence

$$E_n \rightarrow E$$

in E_{reg} , one requires

$$\Phi(E_n) \rightarrow \Phi(E).$$

Meaning

Small variations in the emergent structure cannot produce arbitrary jumps in the observable configuration.

Axiom P3 — Compressivity

There exists a constant

$$0 < \kappa < 1$$

such that

$$\mathcal{I}(\Phi(E)) \leq \kappa \mathcal{C}(E).$$

Interpretation

The projection preserves only a fraction of the available structural information. This formalizes the reductive nature of projection.

Axiom P4 — Non-Invertibility

In general, no map

$$\Phi^{-1}$$

exists on all of $\Phi(E_{\text{reg}})$.

Meaning

Different emergent structures may produce the same projected configuration. Projection entails loss of information.

2.5 Class of Admissible Operators

Definition 2.4

Let $\mathfrak{P}$ denote the class of all operators Φ satisfying axioms P1–P4.

Theorem 2.1 — Non-Emptiness of the Class

The class $\mathfrak{P}$ is non-empty.

Proof

Consider the function

$$\Phi_0(E) = \kappa \mathcal{C}(E).$$

It depends only on global complexity.

Since:

- $\mathcal{C}(E) \geq 0$;
- $\mathcal{C}$ is continuous;
- $0 < \kappa < 1$;

it follows that:

- Φ_0 is continuous;
- Φ_0 is monotonic;
- Φ_0 is compressive;
- Φ_0 is non-invertible, since many configurations have the same complexity.

Therefore:

$$\Phi_0 \in \mathfrak{P}.$$

Observation

This theorem is important. For the first time, the PSE possesses an explicit construction of a projection operator. It is not yet the physically relevant operator, but it demonstrates that the theory is not formally empty.

2.6 Projection Entropy

The absence of invertibility suggests introducing a new quantity.

Definition 2.5

Let

$$p = \Phi(E).$$

Define the *projection fiber*:

$$\mathcal{F}(p) = \{E \in E_{\text{reg}} : \Phi(E) = p\}.$$

Definition 2.6

Define the *projection entropy*:

$$S_\Phi(p) = \log |\mathcal{F}(p)|.$$

Interpretation

Projection entropy measures how many emergent structures become indistinguishable after projection. It directly quantifies the information loss introduced by Φ .

2.7 Stability of Operators

Theorem 2.2 — Uniform Stability

Let

$$\Phi \in \mathfrak{P}.$$

Then there exists a constant

$$L > 0$$

such that

$$d_{\mathcal{P}}(\Phi(E_1), \Phi(E_2)) \leq L \|E_1 - E_2\|_{E_{\text{reg}}}$$

for sufficiently small perturbations.

Proof

It follows from the continuity of the operator and the completeness of the spaces involved.

Conclusion of the First Part

We have obtained:

- a rigorous definition of projection operator;

- an explicit mathematical class of admissible operators;
- an existence theorem;
- a measure of informational loss;
- a first stability result.

Chapter 3 — Hierarchy of Projection Operators

3.1 Projection Complexity

Motivation

The existence of a critical threshold does not automatically imply the appearance of a new projection class. For a projection to be definable, the emergent structure must possess a minimum complexity sufficient to support it.

We therefore introduce a measure of the intrinsic complexity required by a projection operator.

Definition 3.1 — Intrinsic Projection Complexity

Let

$$\Phi \in \mathfrak{P}.$$

The intrinsic complexity of Φ is defined as the quantity

$$\kappa(\Phi) = \inf\{ C(E) : \Phi(E) \text{ is well defined} \}.$$

Interpretation

The quantity $\kappa(\Phi)$ measures the minimum structural organization required for the projection to exist.

Definition 3.2 — Projection Class

Two operators Φ_a, Φ_b are said to be equivalent if they possess the same intrinsic complexity. The equivalence class associated with the level Θ_k is denoted by

$$[\Phi_k].$$

Theorem 3.1

For every critical value Θ_k , there exists a projection class $[\Phi_k]$ defined by

$$\kappa(\Phi_k) = \Theta_k.$$

Proof

This follows directly from the definition of critical value and the definition of intrinsic complexity.

3.2 Hierarchy of Projection Classes

Motivation

The theory developed so far shows that every projection operator possesses an intrinsic complexity $\kappa(\Phi)$. The critical complexity values

$$\Theta_1 < \Theta_2 < \dots < \Theta_n$$

therefore allow us to classify operators into structurally distinct families.

The resulting hierarchy constitutes the mathematical origin of the projection classes of the PSE.

Definition 3.3 — Complexity Class

For each critical value Θ_k , define the class

$$\mathfrak{P}_k = \{\Phi \in \mathfrak{P}: \Theta_k \leq \kappa(\Phi) < \Theta_{k+1}\}.$$

For the last threshold, set:

$$\mathfrak{P}_n = \{\Phi: \kappa(\Phi) \geq \Theta_n\}.$$

Interpretation

The class $\mathfrak{P}_k$ contains all operators that require a level of complexity compatible with the structural regime between two consecutive thresholds.

Theorem 3.2 — Partition of the Operator Class

The classes

$$\mathfrak{P}_1, \mathfrak{P}_2, \dots, \mathfrak{P}_n$$

form a partition of $\mathfrak{P}$.

Proof

The intervals $[\Theta_k, \Theta_{k+1})$ are disjoint. Every operator possesses an intrinsic complexity $\kappa(\Phi)$. Therefore, it belongs to one and only one class.

Thus:

$$\mathfrak{P} = \bigcup_{k=1}^n \mathfrak{P}_k.$$

Definition 3.4 — Minimal Operators

A *minimal operator* of the class $\mathfrak{P}_k$ is an operator Φ_k such that

$$\kappa(\Phi_k) = \Theta_k.$$

Interpretation

The operators

$$\Phi_1, \Phi_2, \dots, \Phi_n$$

represent the structurally simplest configurations capable of existing at their respective complexity levels. They constitute the generating elements of the hierarchy.

Theorem 3.3 — Ordering of the Classes

For the minimal operators:

$$\kappa(\Phi_1) < \kappa(\Phi_2) < \dots < \kappa(\Phi_n).$$

Proof

This follows immediately from

$$\kappa(\Phi_k) = \Theta_k$$

and from the ordering of the critical thresholds:

$$\Theta_1 < \Theta_2 < \dots < \Theta_n.$$

First Principle of Projection Emergence

Theorem 3.4

Let E be an emergent state. If

$$C(E) < \Theta_k,$$

then no operator belonging to $\mathfrak{P}_k$ can be defined on E .

Proof

By definition:

$$\kappa(\Phi) = \inf\{C(E) : \Phi(E) \text{ defined}\}.$$

If

$$C(E) < \Theta_k = \kappa(\Phi_k),$$

the available complexity does not reach the minimum required by the operator. Therefore, the projection cannot be constructed.

3.3 Representational Capacity of Operators

Motivation

Intrinsic complexity $\kappa(\Phi)$ measures the minimum organizational cost required for the existence of a projection. However, it does not describe what the projection is actually capable of representing.

We therefore introduce a measure of the representational power of an operator.

Definition 3.5 — Representational Capacity

Let

$$\Phi \in \mathfrak{P}.$$

The representational capacity of Φ is defined as:

$$R(\Phi) = \dim(\Phi(E_{\text{reg}})),$$

where $\Phi(E_{\text{reg}})$ denotes the image of the operator.

Interpretation

The quantity $R(\Phi)$ measures the number of independent degrees of freedom that can survive the projection. A more sophisticated projection can preserve a richer structure.

Definition 3.6 — Effective Complexity

The effective complexity of the operator Φ is defined as:

$$\Gamma(\Phi) = \kappa(\Phi) R(\Phi).$$

Interpretation

- Intrinsic complexity measures the minimum structural cost.
- Representational capacity measures the richness of the image.

The quantity $\Gamma(\Phi)$ combines both aspects.

Theorem 3.5 — Monotonicity of Capacity

Let Φ_a, Φ_b be two operators such that

$$\kappa(\Phi_a) < \kappa(\Phi_b).$$

Then, for a coherent hierarchy:

$$R(\Phi_a) \leq R(\Phi_b).$$

Proof

Suppose by contradiction that

$$R(\Phi_a) > R(\Phi_b).$$

The less complex operator would then require less structural organization while representing more information. This contradicts the compressivity principle and the definition of intrinsic complexity.

Therefore:

$$R(\Phi_a) \leq R(\Phi_b).$$

Corollary 3.1

For the minimal operators:

$$R(\Phi_1) \leq R(\Phi_2) \leq \dots \leq R(\Phi_n).$$

3.4 Projection Levels

Definition 3.7 — Projection Level

The projection level $L(\Phi)$ is defined as the index of the class to which the operator belongs. In particular:

$$L(\Phi) = k \text{ if } \Phi \in \mathfrak{P}_k.$$

Interpretation

The level does not measure a physical quantity. It represents the degree of structural sophistication of the projection.

Theorem 3.6 — Hierarchical Emergence

If

$$C(E) < \Theta_k,$$

then

$$L(\Phi) < k$$

for every projection definable on E .

Proof

From Theorem 3.4, no projection of class $\mathfrak{P}_k$ can be constructed. It follows that any definable operator must belong to a lower class.

Principle of Structural Saturation

Theorem 3.7

Let $E(\lambda)$ be a continuous family of emergent states. If

$$C(E(\lambda))$$

exceeds a critical threshold Θ_k , then at least one new class of admissible projection operators appears.

Proof

The threshold Θ_k is defined as the minimum complexity required by a new class. Once this threshold is exceeded, the family of definable operators strictly expands. Therefore, at least one new projection class appears.

Conclusion

At this point we have obtained a very important result:

$$C(E) \rightarrow \Theta_k \rightarrow \mathfrak{P}_k$$

that is:

- complexity generates thresholds;
- thresholds generate projection classes;
- projection classes generate representational capacities.

Chapter 4 — Constructive Construction of Projection Operators

4.1 Motivation

The theory developed in the previous chapters has demonstrated:

- the existence of emergent complexity;
- the existence of critical thresholds;
- the existence of projection classes.

A fundamental question remains open:

how is a projection operator concretely constructed?

A constructive theory requires exhibiting an explicit family of operators that satisfies the axioms:

- monotonicity;
- continuity;
- compressivity;
- non-invertibility.

This section provides a general construction of such operators.

4.2 Structural Compression Operators

Definition 4.1

Let

$$E = (D, C) \in E_{\text{reg}}.$$

Define a *structural characteristic function*

$$\Psi(E) = (u_1(E), u_2(E) \dots u_m(E))$$

where each functional u_i is continuous and invariant under isomorphisms.

Examples

Possible choices include:

$$u_1 = C_D, u_2 = C_C, u_3 = C_T,$$

or additional emergent invariants.

Interpretation

The function Ψ extracts only part of the information contained in the emergent structure. It represents the first stage of the projection.

Definition 4.2

Define a *compression operator*

$$K: \mathbb{R}^m \rightarrow \mathcal{P}$$

continuous.

The associated projection operator is:

$$\Phi = K \circ \Psi.$$

Theorem 4.1 — Continuity

The operator $\Phi = K \circ \Psi$ is continuous.

Proof

The functionals u_i are continuous. Therefore, Ψ is continuous. Since K is continuous, the composition of continuous maps is continuous. Thus, Φ is continuous.

4.3 Linear Class

The simplest construction is obtained by choosing

$$\mathcal{P} = \mathbb{R}.$$

Definition 4.3

Define:

$$\Phi_L(E) = \sum_{i=1}^m a_i u_i(E)$$

with $a_i \geq 0$.

Theorem 4.2 — Monotonicity

If $a_i \geq 0$, then Φ_L is monotonic with respect to complexity.

Proof

Any increase in the structural invariants produces an increase in the image. Therefore:

$$C(E_1) \leq C(E_2) \Rightarrow \Phi_L(E_1) \leq \Phi_L(E_2).$$

4.4 Non-Linear Class

Realistic projections are generally not linear. We therefore introduce a more general family.

Definition 4.4

Let

$$F: \mathbb{R}^m \rightarrow \mathbb{R}$$

be a continuous and monotonic function.

Define:

$$\Phi_F(E) = F(u_1(E), \dots, u_m(E)).$$

Canonical Example

$$\Phi_F(E) = \log \left(1 + \sum_{i=1}^m a_i u_i(E) \right).$$

Meaning

This choice naturally realizes:

- saturation;
- compression;
- informational loss.

All desirable properties for a PSE projection.

4.5 Non-Invertibility Theorem

Theorem 4.3

The family Φ_F is not invertible in general.

Proof

The dimension of the emergent space E_{reg} is infinite. The image $\Phi_F(E)$ depends only on a finite number of invariants. Therefore, there exist infinitely many states $E_1, E_2, \dots$ that produce the same value.

Thus:

$$\Phi_F(E_1) = \Phi_F(E_2) \text{ while } E_1 \neq E_2.$$

The map is not invertible.

4.6 Fundamental Operator of the PSE

Definition 4.5

Define the *fundamental operator*:

$$\Phi_*(E) = \log (1 + \alpha_D C_D + \alpha_C C_C + \alpha_T C_T).$$

Since

$$C(E) = \alpha_D C_D + \alpha_C C_C + \alpha_T C_T,$$

we obtain:

$$\boxed{\Phi_*(E) = \log (1 + C(E))}$$

Theorem 4.4 — Properties of the Fundamental Operator

The operator Φ_* satisfies:

- continuity;
- monotonicity;
- compressivity;
- non-invertibility.

Proof

A direct consequence of Theorems 1.10, 2.1, and 4.3, and of the monotonicity of the logarithm.

Fundamental Result

At this point, the PSE finally possesses an object it previously lacked:

$$\boxed{\Phi_*(E) = \log (1 + C(E))}$$

This is no longer an abstract symbol, but an explicit mathematical construction derived from complexity theory.

It represents the first concrete candidate for an *emergent projection* and constitutes the mathematical bridge between:

$$E_{\text{reg}} \rightarrow C(E) \rightarrow \Phi_*(E).$$

Part III — Emerging Geometry and Dynamics

Chapter 5 — Emerging Geometry of Projection

5.1 Motivation

The fundamental operator

$$\Phi_*(E) = \log(1 + C(E))$$

provides an observable quantity derived from the emergent structure.

However, a physical theory cannot be constructed solely from a global scalar. It is necessary to introduce a set of geometric variables capable of describing *local* configurations of the projection.

The objective of this chapter is to show how a geometric structure can naturally emerge from projection operators.

5.2 Emerging Coordinates

Definition 5.1

Let

$$E = (D, C) \in E_{\text{reg}}.$$

Let

$$\{u_i\}_{i=1}^m$$

be a family of structural invariants.

Define the *emergent coordinate map*

$$X_E: E_{\text{reg}} \rightarrow \mathbb{R}^m$$

by

$$X_E(E) = (u_1(E), u_2(E) \dots u_m(E)).$$

Interpretation

Emergent coordinates do not yet represent spatial coordinates. They represent the “position” of the emergent structure in the space of its invariants.

Definition 5.2

The image

$$\mathcal{M}_E = X_E(E_{\text{reg}})$$

is called the *emergent manifold*.

Theorem 5.1 — Existence of the Emergent Manifold

If the functionals u_i are continuous, then

$$\mathcal{M}_E$$

is a separable topological space.

Proof

The coordinate maps are continuous. The continuous image of a metrizable space inherits a natural topological structure. Therefore, $\mathcal{M}_E$ is well defined.

5.3 Induced Emergent Distance

Once an emergent manifold is obtained, we can introduce a notion of distance.

Definition 5.3

For two emergent states E_1, E_2 , define

$$d_E(E_1, E_2) = \sqrt{\sum_{i=1}^m (u_i(E_1) - u_i(E_2))^2}.$$

Interpretation

Two states are close if they possess similar structural invariants. The notion of proximity therefore *emerges* from the structure and is not postulated.

Theorem 5.2

The function d_E defines a metric on $\mathcal{M}_E$.

Proof

This follows immediately from the properties of the Euclidean metric.

5.4 Emerging Information Tensor

To construct a differential geometry, a local object must be introduced.

Definition 5.4

Define the *emergent metric tensor*:

$$g_{ij} = \frac{\partial u_i}{\partial \xi^\alpha} \frac{\partial u_j}{\partial \xi^\alpha}$$

where ξ^α are local coordinates on the emergent manifold.

Interpretation

The tensor g_{ij} measures the sensitivity of the structure with respect to infinitesimal variations of the invariants.

Theorem 5.3

If the derivatives exist and have maximal rank, then g_{ij} is positive semidefinite.

Proof

For any vector $v = (v^1, \dots, v^m)$,

$$v^i g_{ij} v^j = \sum_{\alpha} \left(v^i \frac{\partial u_i}{\partial \xi^{\alpha}} \right)^2 \geq 0.$$

5.5 Principle of Geometric Emergence

Theorem 5.4

Every family of projection operators compatible with the PSE naturally induces a geometric structure

$$(\mathcal{M}_E, g).$$

Proof

The invariants define coordinates. The coordinates define a manifold. Their derivatives induce a metric tensor. Therefore, every projection generates an emergent geometry.

Chapter 6 — Dynamics of Emerging Complexity

6.1 Evolutionary Families

Definition 6.1

Consider a continuous family of emergent structures

$$E(\lambda) = (D_{\lambda}, C_{\lambda})$$

parametrized by $\lambda \in I$.

Assume:

$$D_{\lambda} \text{ and } C_{\lambda}$$

are continuous with respect to λ .

Interpretation

The parameter λ is not yet physical time. It represents an abstract parameter of structural evolution.

Definition 6.2

The trajectory of the emergent structure is

$$\gamma: I \rightarrow E_{\text{reg}}$$

defined by

$$\gamma(\lambda) = E(\lambda).$$

6.2 Complexity Velocity

Definition 6.3

Define the *complexity velocity*:

$$V_C(\lambda) = \frac{d}{d\lambda} C(E(\lambda)).$$

Interpretation

The quantity V_C measures the rate at which structural organization increases or decreases.

Definition 6.4

Define the *complexity acceleration*:

$$A_c(\lambda) = \frac{d^2}{d\lambda^2} C(E(\lambda)).$$

6.3 Principle of Minimal Evolution

This is the first genuine dynamical principle.

Dynamical Axiom D1

Among all possible trajectories, the system selects those that minimize the functional

$$\mathcal{A}[\gamma] = \int_I \mathcal{L}(C, \dot{C}) d\lambda.$$

Interpretation

The emergent structure evolves along optimal paths in complexity space. This role is analogous to the principle of least action in theoretical physics.

Definition 6.5

Choose as fundamental Lagrangian:

$$\mathcal{L} = \frac{1}{2} \dot{C}^2 - U(C),$$

where $U(C)$ represents a *complexity potential*.

Theorem 6.1

The extremal trajectories satisfy:

$$\frac{d^2 C}{d\lambda^2} + U'(C) = 0.$$

Proof

Applying the Euler–Lagrange equations to the functional

$$\mathcal{A} = \int \left(\frac{1}{2} \dot{C}^2 - U(C) \right) d\lambda,$$

we obtain:

$$\frac{d}{d\lambda} \frac{\partial \mathcal{L}}{\partial \dot{C}} - \frac{\partial \mathcal{L}}{\partial C} = 0.$$

Since:

$$\frac{\partial \mathcal{L}}{\partial \dot{C}} = \dot{C}, \quad \frac{\partial \mathcal{L}}{\partial C} = -U'(C),$$

it follows immediately that:

$$\ddot{C} + U'(C) = 0.$$

6.4 Stability of Thresholds

The critical thresholds introduced in Chapter 1 must now be reinterpreted dynamically.

Definition 6.6

A threshold Θ_k is said to be *stable* if:

$$U'(\Theta_k) = 0, U''(\Theta_k) > 0.$$

Interpretation

Thresholds become stable points of the complexity dynamics. They are not merely critical values: they are *structural attractors* of emergent evolution.

Theorem 6.2

If

$$U''(\Theta_k) > 0,$$

then small perturbations of complexity do not destroy the regime associated with Θ_k .

Proof

This follows from standard linear stability theory. Linearizing:

$$C = \Theta_k + \varepsilon,$$

we obtain:

$$\ddot{\varepsilon} + U''(\Theta_k) \varepsilon = 0.$$

The solutions remain bounded.

Conceptual Result

With this chapter, the PSE ceases to be a purely static theory. A genuine dynamics appears:

$$E \rightarrow C \rightarrow \Theta \rightarrow \Phi \rightarrow \mathcal{M}$$

but also a complexity function

$$C(\lambda)$$

that evolves according to a variational principle.

Part IV – Mathematical Consolidation

Objective

The theory developed in the previous parts has introduced:

- emergent complexity;
- critical thresholds;
- projection operators;
- emergent geometry;
- complexity dynamics.

Although these constructions are formally coherent, some proofs rely on implicit assumptions or on properties not yet explicitly justified.

The purpose of this part is to strengthen the mathematical foundations of the theory through a systematic review of the assumptions used. In particular, we analyze:

- the functional structure of the emergent space;
- the regularity of the complexity functionals;
- topological stability;
- the theory of projection operators;
- emergent geometry;
- complexity dynamics.

The goal is not to modify the theory, but to make explicit the mathematical conditions already implicitly used in the previous chapters.

C1 – Completeness of the Emergent Space

Motivation

The entire PSE theory is built on the structural space

$$E_{\text{reg}} = C^0(X \times X) \times C^0(X \times X).$$

All subsequent definitions depend on treating this space as a stable functional environment. It is therefore necessary to rigorously verify the analytical properties of E_{reg} .

Recall of Assumptions

We assume:

- **H1:** X compact.
- **H2:** $D_{S,\text{reg}}^{(e)}$ and $C_{e,\text{reg}}$ continuous.
- **H3:** $D_{S,\text{reg}}^{(e)}$ and $C_{e,\text{reg}}$ uniformly bounded.

Definition C1.1 — Structural Norm

For every element

$$E = (D, C) \in E_{\text{reg}},$$

define

$$\|E\|_{E_{\text{reg}}} = \max(\|D\|_{\infty}, \|C\|_{\infty}).$$

Proposition C1.1

The function

$$\|\cdot\|_{E_{\text{reg}}}$$

is a norm on E_{reg} .

Proof

We must verify:

Positivity

$$\|E\|_{E_{\text{reg}}} \geq 0.$$

Moreover,

$$\|E\|_{E_{\text{reg}}} = 0$$

implies $\|D\|_{\infty} = 0$ and $\|C\|_{\infty} = 0$, hence $D = 0$ and $C = 0$. Therefore $E = 0$.

Homogeneity For every $\lambda \in \mathbb{R}$,

$$\|\lambda E\|_{E_{\text{reg}}} = |\lambda| \|E\|_{E_{\text{reg}}}.$$

Triangle inequality Since

$$\|D_1 + D_2\|_{\infty} \leq \|D_1\|_{\infty} + \|D_2\|_{\infty},$$

and similarly for C , it follows that

$$\|E_1 + E_2\|_{E_{\text{reg}}} \leq \|E_1\|_{E_{\text{reg}}} + \|E_2\|_{E_{\text{reg}}}.$$

Thus, this is indeed a norm. $\square$

Theorem C1.1 — Completeness of the Emergent Space

The space

$$(E_{\text{reg}}, \|\cdot\|_{E_{\text{reg}}})$$

is complete.

Proof

Since X is compact, the space

$$C^0(X \times X)$$

equipped with the uniform norm is a Banach space.

It follows that any Cauchy sequence (D_n) converges to a continuous function D . The same holds for (C_n) .

Let

$$E_n = (D_n, C_n)$$

be a Cauchy sequence in E_{reg} . The two components converge respectively to D and C .

Therefore,

$$E_n \rightarrow (D, C)$$

in the structural norm.

Thus, E_{reg} is complete.

Corollary C1.1

Every convergent structural sequence has a limit that still belongs to the emergent space.

Meaning

The emergent structure is closed under approximation processes. This result guarantees that all subsequent constructions of the theory:

- complexity;
- thresholds;
- projections;
- emergent geometries;

are defined within a stable functional space.

C2 – Regularity of the Complexity Functionals

Motivation

In Chapter 1, the three fundamental functionals of the theory were introduced:

$$C_D, C_C, C_T.$$

The global complexity function

$$C = \alpha_D C_D + \alpha_C C_C + \alpha_T C_T$$

was constructed as a linear combination of these quantities.

For the theory to be mathematically stable, it is necessary to show that the introduced functionals possess adequate regularity properties. In particular, we must verify:

- boundedness;
- continuity;
- stability under perturbations;
- continuous dependence on emergent data.

C2.1 Regularity of Metric Complexity

Recall the definition:

$$C_D(E) = \int_{X \times X} w_D(x, y) D_{S, \text{reg}}^{(e)}(x, y) d\mu(x) d\mu(y).$$

Theorem C2.1 — Lipschitz Continuity of Metric Complexity

For every

$$E_1 = (D_1, C_1), E_2 = (D_2, C_2),$$

one has:

$$|C_D(E_1) - C_D(E_2)| \leq \|D_1 - D_2\|_\infty.$$

Proof

By definition:

$$C_D(E_1) - C_D(E_2) = \int w_D (D_1 - D_2) d\mu d\mu.$$

Taking absolute values:

$$|C_D(E_1) - C_D(E_2)| \leq \int w_D |D_1 - D_2| d\mu d\mu.$$

Since:

$$|D_1 - D_2| \leq \|D_1 - D_2\|_\infty,$$

we obtain:

$$|C_D(E_1) - C_D(E_2)| \leq \|D_1 - D_2\|_\infty \int w_D d\mu d\mu.$$

Using the normalization:

$$\int w_D d\mu d\mu = 1,$$

we conclude:

$$|C_D(E_1) - C_D(E_2)| \leq \|D_1 - D_2\|_\infty.$$

Corollary C2.1

The functional C_D is Lipschitz continuous.

C2.2 Regularity of Correlational Complexity

Recall:

$$C_C(E) = \int |C_{e, \text{reg}}| d\mu d\mu.$$

Theorem C2.2 — Lipschitz Continuity of Correlational Complexity

For every pair of emergent states E_1, E_2 , one has:

$$|C_C(E_1) - C_C(E_2)| \leq \|C_1 - C_2\|_\infty.$$

Proof

Use the elementary inequality:

$$||a| - |b|| \leq |a - b|.$$

Thus:

$$|C_C(E_1) - C_C(E_2)| = \left| \int (|C_1| - |C_2|) \right|.$$

Applying the triangle inequality:

$$|C_C(E_1) - C_C(E_2)| \leq \int ||C_1| - |C_2|| \, d\mu \, d\mu.$$

Hence:

$$|C_C(E_1) - C_C(E_2)| \leq \int |C_1 - C_2| \, d\mu \, d\mu.$$

Using the uniform norm:

$$|C_C(E_1) - C_C(E_2)| \leq \|C_1 - C_2\|_\infty.$$

Corollary C2.2

The functional C_C is Lipschitz continuous.

C2.3 Stability of Quantitative Components

The previous results can be unified.

Theorem C2.3 — Quantitative Stability

For every pair of emergent states

$$E_1, E_2 \in E_{\text{reg}},$$

one has:

$$|C_D(E_1) - C_D(E_2)| \leq \|E_1 - E_2\|_{E_{\text{reg}}},$$

and

$$|C_C(E_1) - C_C(E_2)| \leq \|E_1 - E_2\|_{E_{\text{reg}}}.$$

Proof

This follows immediately from the definition:

$$\|E_1 - E_2\|_{E_{\text{reg}}} = \max(\|D_1 - D_2\|_\infty, \|C_1 - C_2\|_\infty),$$

and from Theorems C2.1 and C2.2.

C2.4 Consequences for the Theory

The results show that the two fundamental quantitative components of the PSE:

$$C_D, C_C,$$

possess a regularity stronger than mere continuity: they are **Lipschitz continuous**.

This implies that:

- small variations of the emergent structure generate small variations of complexity;
- spontaneous divergences cannot appear;
- quantitative estimates of the theory are stable.

In particular, the mathematical structure of the PSE is robust with respect to approximation errors.

Conclusion of Consolidation 2

We have shown that:

$$C_D \text{ and } C_C$$

are Lipschitz continuous functionals on the emergent space.

C3 – Topological Stability and Regularity of Topological Complexity

Motivation

In Chapter 1, topological complexity was defined through the functional

$$C_T(E_{\text{reg}}) = \sum_{k=0}^{\infty} \omega_k \beta_k,$$

where β_k are the Betti numbers of the simplicial complex associated with the emergent structure.

Unlike metric complexity and correlational complexity, topological complexity depends on *discrete invariants*. Consequently, the behavior of C_T cannot be deduced directly from the continuity properties of

$$D_{S,\text{reg}}^{(e)}, C_{e,\text{reg}}.$$

Indeed, an arbitrarily small perturbation of the emergent correlations may modify the associated graph structure and produce non-infinitesimal variations in the Betti numbers.

It is therefore necessary to identify the conditions under which topological complexity remains stable. This consolidation aims to make such stability mathematically rigorous.

C3.1 Sensitivity of Emerging Topology

Fundamental Observation

The construction of the emergent graph uses an activation threshold

$$\eta > 0.$$

An edge is present if and only if

$$|C_{e,\text{reg}}(x, y)| \geq \eta.$$

Thus, a small variation of $C_{e,\text{reg}}$ may modify the set of edges whenever values lie close to the threshold.

In such situations, the following may appear or disappear:

- connected components;
- cycles;
- cavities;
- higher-order simplices.

Hence, topology has an intrinsically discrete nature.

Definition C3.1 — Correlational Stability Margin

Define the *correlational stability margin*:

$$\delta_\eta = \inf_{(x,y) \in X \times X} \|C_{e,\text{reg}}(x,y) - \eta\|.$$

Interpretation

The quantity δ_η measures the minimal distance between emergent correlations and the activation threshold. The larger this quantity is, the more robust the topological structure becomes.

Proposition C3.1

If $\delta_\eta > 0$, then no emergent correlation coincides with the threshold.

Proof

By definition of infimum,

$$\delta_\eta > 0$$

implies

$$\|C_{e,\text{reg}}(x,y) - \eta\| \neq 0$$

for every $(x,y) \in X \times X$.

Thus, no edge of the emergent graph lies in a critical position with respect to the threshold.

C3.2 Homological Stability

Definition C3.2 — Topologically Stable Region

The emergent domain E_{reg} belongs to a *topologically stable region* if

$$\delta_\eta > 0.$$

Interpretation

In a topologically stable region, there exists a finite separation between all emergent correlations and the activation threshold. This prevents sudden changes in the combinatorial structure under sufficiently small perturbations.

Theorem C3.1 — Invariance of the Emergent Graph

Let E_{reg} be topologically stable. If $\tilde{C}_{e,\text{reg}}$ satisfies

$$\|\tilde{C}_{e,\text{reg}} - C_{e,\text{reg}}\|_\infty < \frac{\delta_\eta}{2},$$

then the emergent graph associated with $\tilde{C}_{e,\text{reg}}$ coincides with the graph associated with $C_{e,\text{reg}}$.

Proof

For every pair (x, y) ,

$$|\tilde{C}_{e,\text{reg}}(x, y) - C_{e,\text{reg}}(x, y)| < \frac{\delta_\eta}{2}.$$

Since each value $|C_{e,\text{reg}}(x, y)|$ lies at a distance greater than δ_η from the threshold, a perturbation smaller than $\delta_\eta/2$ cannot cross the level η .

Thus, the presence or absence of each edge remains unchanged. Therefore, the two graphs coincide.

Corollary C3.1

In topologically stable regions, the simplicial complex K_η remains invariant under sufficiently small perturbations.

Proof

The simplicial complex depends solely on the emergent graph. By Theorem C3.1, the graph does not change. Hence, the complex does not change either.

C3.3 Regularity of Topological Complexity

Theorem C3.2 — Local Invariance of Betti Numbers

In topologically stable regions, all Betti numbers are locally constant.

Proof

By Corollary C3.1, the simplicial complex remains invariant. Since homology groups depend exclusively on the complex,

$$H_k(\tilde{K}_\eta) = H_k(K_\eta),$$

and therefore

$$\tilde{\beta}_k = \beta_k.$$

Theorem C3.3 — Local Continuity of Topological Complexity

In topologically stable regions, the functional C_T is locally constant.

Proof

By Theorem C3.2, all Betti numbers remain invariant. Since

$$C_T = \sum_{k=0}^{\infty} \omega_k \beta_k,$$

each term of the series remains constant. Thus, C_T remains constant in a suitable neighborhood of the emergent state.

Corollary C3.2

In topologically stable regions, the functional C_T is continuous.

Proof

Every locally constant function is continuous. Applying Theorem C3.3 yields the result.

C3.4 Critical Regimes

Definition C3.3 — Topologically Critical State

An emergent state is *topologically critical* if

$$\delta_\eta = 0.$$

Interpretation

In a critical state, emergent correlations exist arbitrarily close to the activation threshold. Under such conditions, an infinitesimal perturbation may modify:

- graph connectivity;
- the simplicial complex;
- Betti numbers;
- topological complexity.

Theorem C3.4 — Possible Structural Transitions

Every topological transition of the emergent structure can occur only through topologically critical states.

Proof

In a topologically stable region, Theorem C3.1 guarantees invariance of the emergent graph. Consequently, the simplicial complex and Betti numbers remain invariant. Thus, no topological transition is possible.

Therefore, a variation of Betti numbers can occur only when

$$\delta_\eta = 0.$$

Conclusion of Consolidation 3

We have obtained three fundamental results:

- topological complexity is not generally continuous on the entire emergent space;
- it becomes continuous within topologically stable regions;
- topological transitions can occur only through critical states.

C4 – Rigorous Foundations of the Theory of Projection Operators

Motivation

In Chapter 2, the projection operators

$$\Phi: E_{\text{reg}} \rightarrow \mathcal{P}$$

and the class of admissible operators

$$P$$

were introduced.

The theory established the existence of such operators and their fundamental properties. However, several proofs implicitly rely on regularity assumptions that have not yet been formalized.

In particular, it is necessary to clarify:

- the continuity conditions of the operators;
- stability under perturbations;
- the nature of informational compression;
- the structure of projection fibers;
- the relationship between complexity and representation.

The purpose of this consolidation is to make these aspects rigorous.

C4.1 Regularity of Projection Operators

Definition C4.1 — Uniformly Continuous Operator

A projection operator

$$\Phi: E_{\text{reg}} \rightarrow \mathcal{P}$$

is said to be *uniformly continuous* if for every $\varepsilon > 0$ there exists $\delta > 0$ such that

$$\|E_1 - E_2\|_{E_{\text{reg}}} < \delta \Rightarrow d_{\mathcal{P}}(\Phi(E_1), \Phi(E_2)) < \varepsilon.$$

Observation

Uniform continuity is stronger than ordinary continuity. It guarantees that the sensitivity of the projection remains controlled across the entire emergent space.

Theorem C4.1 — Uniform Continuity of Constructive Projections

Let

$$\Phi_F(E) = F(u_1(E), \dots, u_m(E)),$$

with F Lipschitz continuous and each u_i Lipschitz continuous. Then Φ_F is Lipschitz continuous.

Proof

Let L_F be such that

$$\|F(x) - F(y)\| \leq L_F \|x - y\|.$$

For each functional u_i , let L_i satisfy

$$\|u_i(E_1) - u_i(E_2)\| \leq L_i \|E_1 - E_2\|_{E_{\text{reg}}}.$$

Combining the two estimates yields

$$\|\Phi_F(E_1) - \Phi_F(E_2)\| \leq L \|E_1 - E_2\|_{E_{\text{reg}}},$$

for a suitable constant $L > 0$.

Thus, Φ_F is Lipschitz continuous.

Corollary C4.1

Every operator of the form

$$\Phi_*(E) = \log(1 + C(E))$$

is continuous.

Proof

The function $x \mapsto \log(1 + x)$ is continuous for $x \geq 0$. Since $C(E)$ is continuous by the results of Consolidations 2 and 3, the composition is continuous.

C4.2 Stability of Projections

Motivation

A physically meaningful theory cannot allow arbitrarily large changes in the observable representation to arise from arbitrarily small variations of the emergent structure.

Definition C4.2 — Local Stability

A projection operator Φ is *locally stable* if there exist constants $L > 0$ and $r > 0$ such that

$$\|E_1 - E_2\|_{E_{\text{reg}}} < r \Rightarrow d_{\mathcal{P}}(\Phi(E_1), \Phi(E_2)) \leq L \|E_1 - E_2\|_{E_{\text{reg}}}.$$

Theorem C4.2

Every Lipschitz continuous operator is locally stable.

Proof

This follows immediately from the definition of Lipschitz continuity.

C4.3 Structure of Projection Fibers

Motivation

Non-invertibility is one of the essential aspects of the theory. It is necessary to describe its mathematical structure more precisely.

Definition C4.3 — Projection Fiber

For each point $p \in P$, define the fiber

$$\mathcal{F}(p) = \{E \in E_{\text{reg}} : \Phi(E) = p\}.$$

Interpretation

The fiber collects all emergent states that become indistinguishable after projection.

Theorem C4.3

Let Φ_F be a projection depending only on a finite number of structural invariants. Then the projection fibers are, in general, infinite-dimensional.

Proof

The space E_{reg} has infinite dimension. The image of the projection depends only on the finite vector

$$(u_1, \dots, u_m).$$

Constraining a finite number of invariants does not uniquely determine an element of an infinite-dimensional space. Thus, infinitely many emergent states are compatible with the same projection value.

Therefore, the fiber is generally infinite-dimensional.

Corollary C4.2

The loss of information introduced by projection is not accidental but structural.

Meaning

Non-invertibility does not arise from a particular choice of operator. It follows directly from the difference in complexity between the emergent space and the projected representation.

C4.4 Compatibility with Complexity Classes

Theorem C4.4

Let

$$\Phi \in \mathfrak{P}_k.$$

If

$$C(E) < \Theta_k,$$

then $\Phi(E)$ cannot be defined.

Proof

From Chapter 3:

$$\kappa(\Phi) = \Theta_k.$$

By definition of intrinsic complexity, the operator requires at least this level of organization. If

$$C(E) < \Theta_k,$$

the emergent structure does not possess sufficient complexity. Thus, the projection is not admissible.

Theorem C4.5 — Growth of Representational Capacity

Let

$$\Phi_a \in \mathfrak{P}_i, \Phi_b \in \mathfrak{P}_j,$$

with $i < j$.

Assuming the representational coherence principle, one has:

$$R(\Phi_a) \leq R(\Phi_b).$$

Proof

This follows directly from the representational coherence assumption introduced in Chapter 3.

Critical Observation

In the original manuscript, this property was presented as a theorem. From a mathematical standpoint, however, it depends on a specific structural assumption of the theory. It is therefore more appropriate to treat it as a fundamental principle rather than as a purely deductive consequence.

Conclusion of Consolidation 4

We have obtained four fundamental results:

- constructive projection operators are continuous and stable;
- non-invertibility is a structural consequence of the dimensional gap between emergent space and projected space;
- projection fibers are generally infinite in cardinality;
- the hierarchy of projection classes remains coherent with the complexity theory developed in previous chapters.

C5 – Rigorous Foundations of Emerging Geometry

Motivation

In Chapter 5, the emerging geometry associated with projection was introduced. In particular, the following objects were defined:

- $\mathcal{M}_E$, the space of emerging configurations,
- g_{ij} , the emerging metric tensor.

These constructions are conceptually natural, but they require further mathematical formalization. In particular, it is necessary to clarify:

- the differentiable nature of the emerging manifold;
- the immersion conditions in the space of invariants;
- the existence of the metric tensor;
- the fundamental geometric properties;
- the stability of the geometry under structural perturbations.

This consolidation provides such results.

C5.1 Differentiable Structure of the Emerging Manifold

Recall

The emerging manifold was defined as

$$\mathcal{M}_E = X_E(E_{\text{reg}}) \subseteq \mathbb{R}^m,$$

where

$$X_E(E) = (u_1(E), u_2(E) \dots u_m(E)).$$

Hypothesis G1

The structural invariants

$$u_i: E_{\text{reg}} \rightarrow \mathbb{R}$$

are assumed to be of class $\mathcal{C}^1$.

Definition C5.1 — Emergent Regular Map

The map X_E is said to be *emergently regular* if its differential

$$dX_E(E)$$

has constant rank at every point.

Interpretation

This condition prevents local degenerations of the geometric representation. It is the analogue of the regularity condition used in the theory of differentiable manifolds.

Theorem C5.1 — Immersed Manifold Structure

Under Hypothesis G1, if X_E has constant rank r , then

$$\mathcal{M}_E$$

is an immersed differentiable manifold of dimension r .

Proof

By the constant rank theorem, the image of a differentiable map of constant rank is locally diffeomorphic to a Euclidean space of dimension equal to the rank. Therefore, $\mathcal{M}_E$ inherits a natural structure of immersed differentiable manifold.

Corollary C5.1

The coordinates $(u_1, \dots, u_m)$ form a local system of emerging coordinates.

Meaning

The geometry of the PSE does not require coordinates introduced a priori. Coordinates emerge directly from the structural invariants.

C5.2 Emerging Metric

Motivation

A differentiable manifold does not automatically possess a metric geometry. It is necessary to introduce a bilinear form that allows one to measure lengths and distances.

Definition C5.2 — Emerging Metric Tensor

Let ξ^α be local coordinates on the emerging manifold. Define:

$$g_{ij} = \sum_{\alpha} \frac{\partial u_i}{\partial \xi^\alpha} \frac{\partial u_j}{\partial \xi^\alpha}.$$

Interpretation

The tensor g_{ij} measures how the emerging invariants vary locally. It represents the geometric sensitivity of the emergent structure.

Theorem C5.2 — Symmetry of the Emerging Metric

The tensor g_{ij} is symmetric.

Proof

By definition:

$$g_{ij} = \sum_{\alpha} \frac{\partial u_i}{\partial \xi^{\alpha}} \frac{\partial u_j}{\partial \xi^{\alpha}}.$$

Exchanging indices:

$$g_{ji} = \sum_{\alpha} \frac{\partial u_j}{\partial \xi^{\alpha}} \frac{\partial u_i}{\partial \xi^{\alpha}}.$$

The two expressions coincide. Thus $g_{ij} = g_{ji}$.

Theorem C5.3 — Semidefinite Positivity

For every tangent vector

$$v = (v^1, \dots, v^m),$$

one has:

$$v^i g_{ij} v^j \geq 0.$$

Proof

We compute:

$$v^i g_{ij} v^j = \sum_{\alpha} \left(\sum_i v^i \frac{\partial u_i}{\partial \xi^{\alpha}} \right)^2.$$

Each term is the square of a real number. Thus each contribution is non-negative, and therefore:

$$v^i g_{ij} v^j \geq 0.$$

Corollary C5.2

The emerging metric tensor defines a Riemannian structure when it has maximal rank.

Proof

In that case, the associated quadratic form is positive definite. Thus it satisfies the axioms of a Riemannian metric.

C5.3 Emerging Geometric Distance

Definition C5.3 — Emerging Length

Let

$$\gamma: [a, b] \rightarrow \mathcal{M}_E$$

be a differentiable curve. Define its length:

$$L(\gamma) = \int_a^b \sqrt{g_{ij} \frac{d\xi^i}{dt} \frac{d\xi^j}{dt}} dt.$$

Definition C5.4 — Intrinsic Emerging Distance

For two points $p, q \in \mathcal{M}_E$, define:

$$d_g(p, q) = \inf_{\gamma} L(\gamma),$$

where the infimum is taken over all curves connecting p and q .

Theorem C5.4 — Coherence of Emerging Distance

If g is Riemannian, then d_g is a metric.

Proof

This follows directly from standard Riemannian geometry.

C5.4 Geometric Stability

Motivation

A physically meaningful emerging geometry must be stable under small deformations of the emergent structure.

Theorem C5.5 — Differential Stability

Let E_1 and E_2 be two emergent states close in the norm

$$\|\cdot\|_{E_{\text{reg}}}.$$

If the invariants u_i are Lipschitz continuous, then the associated metrics

$$g^{(1)}, g^{(2)}$$

differ by a bounded amount.

Proof

Lipschitz continuity of the invariants implies continuity of their derivatives on the domain considered. Consequently, the components of the metric tensor vary continuously. Thus small perturbations of the emergent structure produce small variations in g .

C5.5 Geometric Correspondence Principle

Theorem C5.6

Every admissible projection operator

$$\Phi \in P$$

induces an emerging geometry

$$(\mathcal{M}_E, g).$$

Proof

Every admissible operator depends on a family of structural invariants. The invariants define a coordinate map. The coordinate map generates an immersed manifold. The derivatives of the map induce the metric tensor. Thus every projection operator corresponds to a well-defined emerging geometry.

Conclusion of Consolidation 5

We have obtained the following results:

- the emerging manifold possesses a well-defined differentiable structure;
- the emerging metric tensor is symmetric and semidefinite positive;
- the structure becomes Riemannian under maximal rank conditions;
- the emerging distance is intrinsically defined by the metric;
- the emerging geometry is stable under controlled structural perturbations.

C6 – Rigorous Foundations of the Dynamics of Complexity

Motivation

In Chapter 6, a dynamics of emerging complexity based on the principle of minimal evolution was introduced. In particular, the global complexity function

$$C(E)$$

was treated as a dynamical variable subject to the equation

$$\frac{d^2 C}{d\lambda^2} + U'(C) = 0.$$

Although this construction is formally correct, several mathematical aspects must be clarified:

- the regularity of evolutionary trajectories;
- the existence of the action functional;
- the rigorous derivation of the equations of motion;
- the conservation of dynamical quantities;
- the stability of critical thresholds.

The purpose of this consolidation is to make these elements rigorous.

C6.1 Regularity of Evolutionary Trajectories

Recall

Consider a family of emergent states

$$E(\lambda) = (D_\lambda, C_\lambda)$$

parametrized by $\lambda \in I$.

The associated complexity is

$$\mathcal{C}(\lambda) = \mathcal{C}(E(\lambda)).$$

Hypothesis D1

It is assumed that

$$\mathcal{C}(\lambda) \in \mathcal{C}^2(I).$$

Interpretation

Second-order continuity guarantees the existence of the derivatives required to formulate the dynamical problem.

Definition C6.1 — Admissible Trajectory

An evolutionary trajectory is said to be *admissible* if

$$\mathcal{C} \in \mathcal{C}^2(I).$$

Observation

All variations considered below will be restricted to admissible trajectories.

C6.2 Complexity Action

Definition C6.2 — Action Functional

The action associated with a trajectory $\mathcal{C}(\lambda)$ is defined as

$$\mathcal{A}[\mathcal{C}] = \int_a^b \mathcal{L}(\mathcal{C}, \dot{\mathcal{C}}) d\lambda,$$

where

$$\mathcal{L} = \frac{1}{2} \dot{\mathcal{C}}^2 - U(\mathcal{C}).$$

Interpretation

The first term represents the dynamical content of complexity. The second term represents the structural cost associated with a given level of organization.

Theorem C6.1 — Existence of the Action

If

$$\mathcal{C} \in \mathcal{C}^2(I)$$

and U is continuous, then

$$\mathcal{A}[\mathcal{C}]$$

is well defined.

Proof

The function $\dot{\mathcal{C}}$ is continuous; hence $\dot{\mathcal{C}}^2$ is continuous. Since U is continuous and $\mathcal{C}$ is continuous, the composition $U(\mathcal{C})$ is continuous. Thus the Lagrangian $\mathcal{L}$ is continuous on $[a, b]$. The integral of a continuous function on a compact interval is finite. Therefore, $\mathcal{A}[\mathcal{C}]$ exists.

C6.3 Derivation of the Dynamical Equations

Theorem C6.2 — Euler–Lagrange Equation for Complexity

Stationary trajectories of the action satisfy

$$\frac{d^2\mathcal{C}}{d\lambda^2} + U'(\mathcal{C}) = 0.$$

Proof

From variational theory:

$$\frac{d}{d\lambda} \left(\frac{\partial \mathcal{L}}{\partial \dot{\mathcal{C}}} \right) - \frac{\partial \mathcal{L}}{\partial \mathcal{C}} = 0.$$

Since

$$\mathcal{L} = \frac{1}{2} \dot{\mathcal{C}}^2 - U(\mathcal{C}),$$

we obtain:

$$\frac{\partial \mathcal{L}}{\partial \dot{\mathcal{C}}} = \dot{\mathcal{C}}, \quad \frac{\partial \mathcal{L}}{\partial \mathcal{C}} = -U'(\mathcal{C}).$$

Thus:

$$\ddot{\mathcal{C}} + U'(\mathcal{C}) = 0.$$

C6.4 Conservation of Complexity Energy

Definition C6.3 — Complexity Energy

The energy of the trajectory is defined as

$$E_C^{\text{reg}} = \frac{1}{2} \dot{\mathcal{C}}^2 + U(\mathcal{C}).$$

Interpretation

The complexity energy measures the total dynamical organization present in the trajectory.

Theorem C6.3 — Energy Conservation

If $\mathcal{C}(\lambda)$ satisfies the fundamental dynamical equation, then

$$E_C^{\text{reg}}$$

remains constant.

Proof

Differentiating with respect to λ :

$$\frac{dE_C^{\text{reg}}}{d\lambda} = \dot{\mathcal{C}}\ddot{\mathcal{C}} + U'(\mathcal{C})\dot{\mathcal{C}}.$$

Factorizing:

$$\frac{dE_c^{\text{reg}}}{d\lambda} = \dot{c} (\ddot{c} + U'(c)).$$

Since

$$\ddot{c} + U'(c) = 0,$$

we obtain:

$$\frac{dE_c^{\text{reg}}}{d\lambda} = 0.$$

Thus,

$$E_c^{\text{reg}} = \text{constant}.$$

C6.5 Stability of Critical Thresholds

Definition C6.4 — Dynamical Critical Point

A value θ_k is said to be a *dynamical critical point* if

$$U'(\theta_k) = 0.$$

Interpretation

Thresholds become equilibrium configurations of the complexity dynamics.

Theorem C6.4 — Linear Stability

If

$$U'(\theta_k) = 0 \text{ and } U''(\theta_k) > 0,$$

then θ_k is dynamically stable.

Proof

Let

$$c = \theta_k + \varepsilon.$$

Expanding to first order:

$$U'(c) = U'(\theta_k) + U''(\theta_k)\varepsilon + o(\varepsilon).$$

Since $U'(\theta_k) = 0$, we obtain:

$$\ddot{\varepsilon} + U''(\theta_k)\varepsilon = 0.$$

Because $U''(\theta_k) > 0$, the solutions are oscillatory and bounded. Thus the equilibrium is stable.

Corollary C6.1

Critical thresholds can be interpreted as local attractors of the complexity dynamics.

C6.6 Transition Regimes

Definition C6.5 — Dynamical Transition

The system undergoes a *dynamical transition* when a trajectory moves from the stability basin associated with a threshold θ_i to that associated with a threshold θ_j , with $i \neq j$.

Interpretation

The structural transitions introduced in Chapter 1 now receive a dynamical formulation. They do not represent mere numerical variations of complexity. They correspond to regime changes in the evolution of the emergent structure.

Conclusion of Consolidation 6

We have consolidated the mathematical foundations of complexity dynamics by demonstrating:

- the rigorous existence of the action functional;
- the variational derivation of the fundamental dynamical equation;
- the conservation of complexity energy;
- the linear stability of critical thresholds;
- the interpretation of structural transitions as dynamical regime changes.

C7 – Global Coherence of the Emerging Chain

Motivation

In the previous chapters, the following objects were progressively introduced:

- the emergent space E_{reg} ;
- the complexity function C ;
- the critical thresholds Θ_k ;
- the projection operators Φ ;
- the emerging manifold $\mathcal{M}_E$;
- the metric tensor g ;
- the dynamics of complexity.

Each construction was defined separately, and its main properties were established. It remains to verify that these objects are mutually compatible and form a single mathematical structure. The purpose of this consolidation is to formalize such compatibility.

C7.1 Fundamental Emerging Chain

Definition C7.1

The *emerging chain* of the PSE is defined as the diagram:

$$E_{\text{reg}} \rightarrow \mathbb{R}_+ \rightarrow \{\Theta_k\} \rightarrow \mathcal{P} \rightarrow \mathcal{M}_E \rightarrow (\mathcal{M}_E, g).$$

Interpretation

Each stage of the theory uses only the objects produced by the previous stage. No elements external to the emergent structure are introduced.

C7.2 Functional Compatibility

Theorem C7.1

Every map in the emerging chain is well defined.

Proof

From Consolidation 1, E_{reg} is a Banach space. From Chapter 1 and Consolidation 2, C is well defined and continuous. From Chapter 1, the thresholds Θ_k are obtained as critical values of complexity. From Chapter 2 and Consolidation 4, the operators Φ are well defined. From Chapter 5 and Consolidation 5, the emerging manifold and its associated geometry exist. Therefore, every arrow in the diagram is well defined.

C7.3 Structural Coherence

Definition C7.2

An emergent theory is said to be *structurally coherent* if no subsequent construction requires information not contained in the preceding stages.

Theorem C7.2

The mathematical structure of the PSE is structurally coherent.

Proof

Complexity depends only on the emergent state. Thresholds depend only on complexity. Projections depend only on thresholds and emergent invariants. Geometry emerges from projections. Dynamics acts on complexity. No step introduces new primitive variables. Therefore, the theory is structurally closed.

C7.4 Emerging Closure

Definition C7.3

The PSE satisfies the *principle of emerging closure* if every fundamental quantity of the theory can be traced back to a structure belonging to E_{reg} .

Theorem C7.3

All fundamental quantities of the PSE derive from the emergent space.

Proof

The fundamental quantities are:

$$C, \Theta_k, \Phi, g, E_C^{\text{reg}}.$$

Each of them is constructed through a finite succession of applications starting from elements of E_{reg} . Thus none of the fundamental quantities is independent of the emergent structure.

C7.5 Principle of Mathematical Self-Sufficiency

Theorem C7.4

The advanced mathematical formulation of the PSE is self-sufficient in the sense that every fundamental object of the theory can be defined internally to the system.

Proof

From Theorems C7.1, C7.2, and C7.3 it follows that:

- all fundamental objects exist;
- they are mutually compatible;
- they derive from the same structural starting space.

The theory therefore possesses a closed mathematical formulation.

Conclusion of Consolidation 7

Part IV concludes with a global result. Not only have individual theorems been verified, but the coherence of the entire mathematical architecture has been demonstrated:

$$E_{\text{reg}} \rightarrow C \rightarrow \Theta \rightarrow \Phi \rightarrow \mathcal{M}_E \rightarrow g \rightarrow \text{Dynamics}.$$

Part V – Emergence of the Physical Metric

Objective

The theory developed in the previous parts has allowed us to construct:

- an emergent space E_{reg} ;
- a quantitative theory of complexity;
- a hierarchy of projection operators;
- an emerging manifold $\mathcal{M}_E$;
- an emerging metric g .

However, the metric g is still a purely internal structure of the PSE. It describes relations among emergent invariants, but it does not yet possess a direct physical interpretation. The purpose of this part is to determine the conditions under which the emerging geometry can be identified with a physically observable geometry.

Chapter 7 – Emergence of the Physical Metric

7.1 Geometric Observability Condition

Motivation

Not every emerging geometric structure must necessarily be accessible through projection. A criterion is needed to distinguish purely formal geometries from physically observable ones.

Definition 7.1 – Projectable Geometry

Let $(\mathcal{M}_E, g)$ be an emerging geometry generated by the regularized structure

$$E_{\text{reg}} \in C^0(X \times X) \times C^0(X \times X).$$

The geometry $(\mathcal{M}_E, g)$ is said to be *projectable* if there exists at least one projection operator

$$\Phi \in P$$

such that the metric induced by the projection,

$$g_\Phi := (\Phi^* d_{\mathcal{P}}),$$

coincides locally with g on open sets of $\mathcal{M}_E$.

Interpretation

The projection need not preserve the entire emerging geometry, but it must preserve the essential local metric structure—i.e., the part of the geometry that remains accessible to physical observables.

7.2 Observable Distance

Definition 7.2 – Observable Distance

For $p, q \in \mathcal{M}_E$, define the observable distance:

$$d_{\text{obs}}(p, q) = d_{\mathcal{P}}(\Phi(p), \Phi(q)),$$

where $d_{\mathcal{P}}$ is the metric of the projected space P .

Rigorous Interpretation

The observable distance is the portion of the emerging distance that survives projection. Since Φ may identify distinct points, the observable distance may be degenerate.

Theorem 7.1 – Observable Distance as a Pseudometric

Let

$$d_{\text{obs}}(p, q) = d_{\mathcal{P}}(\Phi(p), \Phi(q)),$$

where $d_{\mathcal{P}}$ is a metric on P and $\Phi: \mathcal{M}_E \rightarrow P$ is a continuous projection operator. Then d_{obs} is a pseudometric on $\mathcal{M}_E$.

Proof

Since $d_{\mathcal{P}}$ is a metric:

- **Non-negativity**

$$d_{\text{obs}}(p, q) = d_{\mathcal{P}}(\Phi(p), \Phi(q)) \geq 0.$$

- **Symmetry**

$$d_{\text{obs}}(p, q) = d_{\mathcal{P}}(\Phi(p), \Phi(q)) = d_{\mathcal{P}}(\Phi(q), \Phi(p)) = d_{\text{obs}}(q, p).$$

- **Identity of indiscernibles**

$$d_{\text{obs}}(p, p) = d_{\mathcal{P}}(\Phi(p), \Phi(p)) = 0.$$

- **Triangle inequality**

$$d_{\text{obs}}(p, r) = d_{\mathcal{P}}(\Phi(p), \Phi(r)) \leq d_{\mathcal{P}}(\Phi(p), \Phi(q)) + d_{\mathcal{P}}(\Phi(q), \Phi(r)) = d_{\text{obs}}(p, q) + d_{\text{obs}}(q, r).$$

- **Possible degeneracy** If $\Phi(p) = \Phi(q)$ with $p \neq q$, then

$$d_{\text{obs}}(p, q) = 0.$$

Thus d_{obs} is not a metric but a pseudometric. The continuity of Φ ensures that d_{obs} depends continuously on emergent states, but is not required for the pseudometric properties.

7.3 Macroscopic Geometric Limit

Motivation

Fundamental emerging structures may possess a large number of degrees of freedom. Physical observation, however, typically occurs at much larger scales. A geometric coarse-graining procedure is therefore required.

Definition 7.3 – Geometric Averaging Operator

Let $L > 0$ be a characteristic scale. Define the geometric averaging operator:

$$\mathcal{G}_L: g \mapsto g^{(L)},$$

where $g^{(L)}$ is obtained by averaging the emerging metric over regions of diameter of order L .

Interpretation

$\mathcal{G}_L$ removes geometric fluctuations associated with scales smaller than L , producing an effective metric at macroscopic scales.

7.4 Emerging Physical Metric

Definition 7.4 – Physical Metric

Define the emerging physical metric:

$$g_{\text{phys}} = \lim_{L \rightarrow \infty} g^{(L)},$$

when such a limit exists and is unique in the uniform topology.

Interpretation

The physical metric is not postulated: it emerges as the macroscopic limit of the geometry generated by projection operators and coarse-graining.

7.5 Theorem 7.2 – Existence of a Limit Geometry

Let $\{g^{(L)}\}_{L>0}$ be a family of emerging metrics defined on a compact domain X , and assume:

- the coefficients of $g^{(L)}$ are uniformly bounded;
- the family is equicontinuous.

Then there exists a subsequence $g^{(L_n)}$ that converges uniformly to a limit metric g_{∞} .

Proof

Consider the coefficients of the metric tensor in a local coordinate system. Each component $g_{ij}^{(L)}$ is a continuous function on a compact set, and the family $\{g_{ij}^{(L)}\}$ is:

- uniformly bounded,
- equicontinuous.

By the Arzelà–Ascoli theorem, every equicontinuous and uniformly bounded family of functions on a compact set admits a uniformly convergent subsequence.

Applying the theorem to each coefficient $g_{ij}^{(L)}$, we obtain a subsequence converging uniformly to limit functions $g_{\infty,ij}$. These functions reconstruct a limit metric tensor g_{∞} . Thus a limit geometry exists.

7.6 Principle of Emergence of Space-Time

Principle E1

The physically observable space-time corresponds to an emerging physical metric

$$g_{\text{phys}}$$

obtained as the macroscopic limit of the geometry generated by projection operators.

Meaning

In this formulation:

- space-time is not fundamental;

- geometry is not postulated;
- the observable metric is a collective effect of the emergent structure.

Chapter 8 – Dynamics of the Emerging Physical Metric

8.1 Motivation

In Chapter 7, the emerging physical metric

$$g_{\text{phys}}$$

was introduced, obtained as the macroscopic limit of the emerging geometry associated with the projection operators. This construction provides an observable geometry, but it does not yet specify the laws governing its evolution. The purpose of this chapter is to determine a natural geometric dynamics for the emerging physical metric.

8.2 Emerging Physical Manifold

Definition 8.1

The *emerging physical space* is defined as the differentiable manifold

$$\mathcal{M}_{\text{phys}}$$

equipped with the metric

$$g_{\text{phys}}.$$

The pair

$$(\mathcal{M}_{\text{phys}}, g_{\text{phys}})$$

is called the *emerging space-time* of the PSE.

Interpretation

Space-time is not assumed as a primitive object. It represents the final result of the projection and coarse-graining process applied to the emergent structure.

8.3 Emerging Curvature

Once the physical metric is introduced, the usual geometric objects can be defined.

Definition 8.2

The Levi-Civita connection associated with g_{phys} is defined through the Christoffel symbols

$$\Gamma_{\mu\nu}^{\alpha}.$$

Definition 8.3

The Riemann curvature tensor is defined as

$$R^{\rho}{}_{\sigma\mu\nu}.$$

Definition 8.4

The Ricci tensor is defined as

$$R_{\mu\nu}.$$

Definition 8.5

The scalar curvature is defined as

$$R.$$

Observation

The differential geometry of the emerging manifold formally coincides with that of standard Riemannian or pseudo-Riemannian manifolds. The fundamental difference lies in the origin of the metric: in the PSE, the metric is not postulated but emerges from the underlying emergent structure.

8.4 Emerging Geometric Action

Motivation

The geometric dynamics must be formulated independently of the coordinate system. For this reason, a scalar geometric functional is introduced.

Definition 8.6

The *emerging geometric action* is defined as

$$\mathcal{A}_g = \int_{\mathcal{M}_{\text{phys}}} \mathcal{L}_g \sqrt{|g|} d^n x,$$

where $\mathcal{L}_g$ is a geometric Lagrangian yet to be determined.

Principle G1

The physically observable dynamics corresponds to the stationary points of the emerging geometric action.

Interpretation

This principle is the geometric analogue of the minimal-evolution principle introduced for complexity.

8.5 Minimal Lagrangian

Definition 8.7

The minimal geometric Lagrangian compatible with:

- invariance under coordinate transformations,
- locality,
- differential dependence of order not higher than second,

is taken to be

$$\mathcal{L}_g = R.$$

Observation

This choice formally coincides with the Hilbert–Einstein Lagrangian. In the PSE, however, this Lagrangian is not introduced as a physical postulate: it emerges as the simplest scalar density constructible from the emerging physical metric.

8.6 Theorem 8.1 – Emerging Geometric Equations

The stationary points of the action

$$\mathcal{A}_g = \int R \sqrt{|g|} d^n x$$

satisfy

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0.$$

Proof

The variation of the action with respect to the metric $g_{\mu\nu}$ yields the associated Euler–Lagrange equations. The derivation coincides with the standard variation of the Hilbert–Einstein action. Thus one obtains

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0.$$

8.7 Emerging Interpretation

Fundamental Result

The geometric equations obtained have the same form as the vacuum Einstein equations. However, the interpretation is radically different.

- In General Relativity, $g_{\mu\nu}$ is a fundamental field.
- In the PSE,

$$g_{\mu\nu} = g_{\text{phys}}$$

is an emergent structure derived from the chain:

$$E_{\text{reg}} \rightarrow \mathcal{C} \rightarrow \Theta \rightarrow \Phi \rightarrow \mathcal{M}_E \rightarrow g \rightarrow g_{\text{phys}}.$$

Chapter 9 – Emergence of Matter and of the Energy–Momentum Tensor

9.1 Motivation

In Chapter 8, the emerging physical metric led to the geometric equations

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = 0.$$

These equations describe a geometry without sources. However, the observed world contains localized structures that, in ordinary physics, are interpreted as matter and energy. The purpose of this chapter is to show how such sources can naturally emerge from the projection theory of the PSE.

9.2 Informational Loss and Projection Residue

Recall

Chapter 4 showed that every projection is non-invertible. For a projected configuration

$$p = \Phi(E),$$

there exists a fiber

$$\mathcal{F}(p) = \{E \in E_{\text{reg}}: \Phi(E) = p\}.$$

Interpretation

An observable configuration does not correspond to a single emergent state. It represents an equivalence class of emergent states that are indistinguishable after projection.

Definition 9.1 – Emerging Degeneracy Density

Define the *degeneracy density* as the function

$$\rho_{\Phi}(p) = \log |\mathcal{F}(p)|.$$

Interpretation

The quantity ρ_{Φ} measures the amount of emergent structure hidden behind a single observable configuration.

9.3 Emerging Geometric Entropy

Definition 9.2

Define the *emerging geometric entropy* as

$$S_g = \int_{\mathcal{M}_{\text{phys}}} \rho_{\Phi}(x) \sqrt{|g|} \, d^n x.$$

Interpretation

This quantity measures the total informational loss introduced by projection. In the PSE, matter is interpreted as the geometric manifestation of the emergent structure that is not fully accessible.

9.4 Emerging Matter Field

Definition 9.3

Define the *emerging scalar field* as

$$\Psi(x) = \rho_{\Phi}(x).$$

Observation

The field Ψ does not yet represent an elementary particle. It locally measures the hidden informational content of the projection.

Definition 9.4

Introduce the emerging Lagrangian

$$\mathcal{L}_{\Psi} = \frac{1}{2} g^{\mu\nu} \nabla_{\mu} \Psi \nabla_{\nu} \Psi - V(\Psi).$$

9.5 Emerging Energy–Momentum Tensor

Definition 9.5

Define

$$T_{\mu\nu} = -\frac{2}{\sqrt{|g|}} \frac{\delta}{\delta g^{\mu\nu}} (\sqrt{|g|} \mathcal{L}_{\Psi}).$$

Theorem 9.1

The tensor $T_{\mu\nu}$ is symmetric.

Proof

This follows directly from the variational definition with respect to the symmetric metric tensor $g_{\mu\nu}$.

Theorem 9.2

If the emerging field equations are satisfied, then

$$\nabla^\mu T_{\mu\nu} = 0.$$

Proof

This follows from the invariance of the action under diffeomorphisms and from the associated Noether identity.

9.6 Geometric Equations with Source

Theorem 9.3

Let

$$\mathcal{A} = \mathcal{A}_g + \mathcal{A}_\Psi$$

be the total action of the system. The stationary points satisfy

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu}.$$

Proof

The variation of the total action with respect to the metric produces two contributions:

- the geometric contribution,
- the emerging contribution.

The sum of the two yields the above equations.

Fundamental Result

For the first time, the term $T_{\mu\nu}$ appears. This tensor is not postulated as fundamental matter. In the PSE, it *emerges* from the hidden structure of the projection fibers.

Schematically:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow \rho_\Phi \rightarrow \Psi \rightarrow T_{\mu\nu}.$$

Part VI – Physical Limits and Correspondence Principle

Objective

An emergent theory can be considered physically acceptable only if it reproduces the established theories in the regimes where such theories have already been experimentally verified. In the case of the PSE, this requirement implies that:

- the emerging geometry must reproduce General Relativity in appropriate limits;
- emerging matter must behave as an ordinary gravitational source;
- observable geometric quantities must coincide with those of known physics.

The purpose of this part is to demonstrate these properties.

Chapter 10 – Relativistic Limit of the PSE

10.1 Motivation

In Chapters 8 and 9, the emerging equations

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu}$$

were obtained. These equations possess the same formal structure as Einstein's equations. It is necessary, however, to demonstrate that General Relativity can be recovered as a limiting case of the PSE.

Definition 10.1 – Stable Macroscopic Regime

An emergent structure is said to belong to the *stable macroscopic regime* if:

- the global complexity varies slowly,

$$\left| \frac{dC}{d\lambda} \right| \ll 1;$$

- the convergent averaged geometry g_{phys} exists;
- projection fluctuations are negligible with respect to the observational scale.

Interpretation

This regime describes the situation ordinarily observed in macroscopic physics. The microscopic components of the emergent structure are averaged out by the coarse-graining procedure.

10.2 Geometric Reduction

Theorem 10.1 – Classical Geometric Limit

In the stable macroscopic regime, the emerging geometry coincides with a pseudo-Riemannian geometry.

Proof

From Chapter 7, the physical metric is defined as the limit

$$g_{\text{phys}} = \lim_{L \rightarrow \infty} g^{(L)}.$$

In the stable macroscopic regime, this limit exists. By construction, the associated connection is the Levi-Civita connection of the limit metric. It follows that the emerging space-time is an ordinary pseudo-Riemannian manifold.

Corollary 10.1

All tools of relativistic differential geometry are applicable to the macroscopic limit of the PSE.

10.3 Reduction of the Emerging Equations

Theorem 10.2 – Einstein–PSE Correspondence

In the stable macroscopic regime, the emerging geometric equations coincide formally with Einstein's equations.

Proof

From Theorem 9.3:

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = \kappa T_{\mu\nu}.$$

This relation coincides with the tensorial structure of the equations of General Relativity. Therefore, the macroscopic limit of the PSE reproduces relativistic geometric dynamics.

Interpretation

General Relativity is not negated by the PSE. It appears as the effective theory of the macroscopic sector of the emergent structure.

10.4 Relativistic Correspondence Principle

Principle CR1

Every prediction of the PSE must reduce to a prediction of General Relativity when the system is in the stable macroscopic regime.

Meaning

This principle plays for the PSE the same role that Bohr's correspondence principle plays in quantum mechanics.

10.5 Interpretation of the Gravitational Field

Fundamental Result

In the PSE, the gravitational field does not represent a fundamental structure. It constitutes the effective description of the emerging geometry observed after projection.

Schematically:

$$E_{\text{reg}} \rightarrow C \rightarrow \Theta \rightarrow \Phi \rightarrow g_{\text{phys}} \rightarrow \text{observable gravity.}$$

Gravity therefore emerges as a collective phenomenon of the emergent structure and not as a primitive entity.

Chapter 11 – Physical Properties of Emerging Matter

11.1 Motivation

In Chapter 9, matter emerged as a consequence of the non-invertibility of the projection. In particular, the chain

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow \rho_\Phi \rightarrow \Psi \rightarrow T_{\mu\nu}$$

was introduced. This construction provides a source for the emerging geometric equations. It remains to verify that this emerging matter possesses physical properties compatible with those required of a gravitational source. The purpose of this chapter is to analyze such properties.

11.2 Emerging Energy Density

Definition 11.1

Let u^μ be a unit observer field. Define the *emerging observed energy density*:

$$\rho_E = T_{\mu\nu} u^\mu u^\nu.$$

Interpretation

The quantity ρ_E measures the energetic content associated with the emergent structure hidden behind the projection.

Theorem 11.1 – Positivity of the Energy Density

If the emerging potential $V(\Psi)$ is non-negative, then

$$\rho_E \geq 0.$$

Proof

From the definition of the Lagrangian,

$$\mathcal{L}_\Psi = \frac{1}{2} g^{\mu\nu} \nabla_\mu \Psi \nabla_\nu \Psi - V(\Psi),$$

the energetic contribution is given by the sum of quadratic terms and the potential. Since

$$(\nabla \Psi)^2 \geq 0, \quad V(\Psi) \geq 0,$$

it follows that

$$\rho_E \geq 0.$$

11.3 Emerging Pressure

Definition 11.2

Define the *emerging isotropic pressure*:

$$P_E = \frac{1}{3} h^{\mu\nu} T_{\mu\nu},$$

where

$$h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$$

is the spatial projector associated with the observer.

Interpretation

The emerging pressure represents the geometric response of projection degeneracies to local variations of the structure.

11.4 Emerging Equation of State

Definition 11.3

Define the *emerging equation-of-state parameter*:

$$w_E = \frac{P_E}{\rho_E}.$$

Interpretation

The quantity w_E allows one to classify different regimes of emerging matter.

Examples

- **Kinetic-dominated regime** If

$$(\nabla\Psi)^2 \gg V(\Psi),$$

then

$$w_E \simeq 1.$$

- **Potential-dominated regime** If

$$V(\Psi) \gg (\nabla\Psi)^2,$$

then

$$w_E \simeq -1.$$

Observation

The mathematical structure of the PSE therefore admits regimes formally analogous to those normally attributed to:

- stiff matter,
- scalar fields,
- vacuum energy.

11.5 Local Conservation

Theorem 11.2 – Conservation of Emerging Matter

If the field equations for Ψ are satisfied, then

$$\nabla^\mu T_{\mu\nu} = 0.$$

Proof

From Theorem 9.2, the energy–momentum tensor derived from the emerging action is conserved. Thus,

$$\nabla^\mu T_{\mu\nu} = 0.$$

Meaning

Emerging matter satisfies the same local conservation law required in General Relativity. This result is essential for ensuring the physical consistency of the theory.

11.6 Energy Conditions

Motivation

A physically acceptable gravitational source must satisfy appropriate energy conditions.

Definition 11.4 – Weak Energy Condition

Emerging matter satisfies the *weak energy condition* if

$$T_{\mu\nu}u^\mu u^\nu \geq 0$$

for every timelike observer.

Theorem 11.3

If $V(\Psi) \geq 0$, then emerging matter satisfies the weak energy condition.

Proof

This follows immediately from Theorem 11.1.

11.7 Emerging Mass

Definition 11.5

Define the *emerging mass* contained in a region $\Omega \subseteq \mathcal{M}_{\text{phys}}$ as

$$M_E(\Omega) = \int_{\Omega} \rho_E \sqrt{|g|} \, d^n x.$$

Interpretation

Emerging mass measures the total amount of structural degeneracy present in a region of the emerging space-time.

Fundamental Result

In the PSE, mass is not a fundamental property. It derives from the amount of structural information that cannot be recovered after projection.

Schematically:

$$E_{\text{reg}} \rightarrow \mathcal{F}(p) \rightarrow \rho_{\Phi} \rightarrow T_{\mu\nu} \rightarrow \rho_E \rightarrow M_E.$$

11.8 Physical Interpretation

The theory now allows matter and energy to be reinterpreted as manifestations of the hidden emergent structure. Within the PSE framework:

- geometry emerges from projection;
- matter emerges from projection degeneracy;
- mass emerges from the distribution of degeneracy;

- gravity emerges from the interaction among these structures.

Chapter 12 – Newtonian Limit of the PSE

12.1 Motivation

In Chapters 8, 9, 10, and 11 it was shown that the PSE leads, in the macroscopic regime, to the geometric equations

$$R_{\mu\nu} - \frac{1}{2}R g_{\mu\nu} = \kappa T_{\mu\nu}.$$

It was also demonstrated that emerging matter possesses properties compatible with a physical gravitational source. It now remains to verify the most stringent requirement of the correspondence principle: **the recovery of Newtonian gravitation.**

Since classical mechanics represents the low-velocity, weak-field limit of General Relativity, a physically acceptable emergent theory must necessarily reproduce this regime as well.

12.2 Weak-Field Regime

Definition 12.1

The system is said to be in the *weak-field regime* if the emerging physical metric can be written as

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

where $\eta_{\mu\nu}$ is the flat Minkowski metric and

$$|h_{\mu\nu}| \ll 1.$$

Interpretation

The observable geometry differs only weakly from flat geometry. This corresponds to gravitational regimes ordinarily observed on astronomical and terrestrial scales.

Definition 12.2

One also assumes the non-relativistic regime:

$$\frac{v}{c} \ll 1,$$

where v is the typical velocity of the emerging sources.

12.3 Emerging Gravitational Potential

Definition 12.3

In the weak-field limit, the *emerging gravitational potential* ϕ is introduced through the relation

$$g_{00} = -\left(1 + \frac{2\phi}{c^2}\right).$$

Interpretation

The quantity ϕ represents the macroscopic gravitational residue produced by the emergent structure. In the PSE it is not a fundamental object: it emerges from the deformation of the physical metric g_{phys} .

12.4 Static Limit

Definition 12.4

An emergent configuration is said to be *static* if

$$\frac{\partial g_{\mu\nu}}{\partial t} = 0, \quad \frac{\partial T_{\mu\nu}}{\partial t} = 0.$$

Interpretation

The emergent structure has reached a stationary macroscopic configuration.

Theorem 12.1 – Reduction of the Energy–Momentum Tensor

In the static and non-relativistic regime,

$$T_{00} \simeq \rho_E c^2.$$

Proof

In the limit $v/c \ll 1$, kinetic and pressure contributions are negligible compared to the dominant energy term. Thus the energy–momentum tensor is dominated by its temporal component:

$$T_{00} \simeq \rho_E c^2.$$

12.5 Emerging Poisson Equation

Theorem 12.2 – Newtonian Limit of the PSE

In the weak-field, static, and non-relativistic regime, the emerging equations of the PSE reduce to

$$\nabla^2 \phi = 4\pi G \rho_E.$$

Proof

In the perturbative limit of the metric

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu},$$

the geometric equations of Chapter 9 take their linearized form. In the static regime, the dominant contribution comes from the 00component.

Using the relation between g_{00} and ϕ , together with the reduction

$$T_{00} \simeq \rho_E c^2,$$

one obtains

$$\nabla^2 \phi = 4\pi G \rho_E.$$

Corollary 12.1

The emerging gravitational field satisfies the Poisson equation.

Meaning

Classical gravitation appears as the low-energy limit of the emergent structure.

12.6 Emerging Gravitational Mass

Definition 12.5

For a localized distribution of emerging matter, define

$$M_E = \int \rho_E(x) d^3x.$$

Interpretation

The observed gravitational mass represents the total structural degeneracy contained in the region considered.

Theorem 12.3 – Potential of a Point Source

For a spherically symmetric localized distribution,

$$\phi(r) = -\frac{GM_E}{r}.$$

Proof

For spherical symmetry, the Poisson equation becomes

$$\frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{d\phi}{dr} \right) = 4\pi G \rho_E.$$

Outside the source, $\rho_E = 0$. The equation reduces to

$$\nabla^2 \phi = 0.$$

The only solution regular at infinity is

$$\phi(r) = -\frac{GM_E}{r}.$$

Corollary 12.2 – Newton's Law

The emerging gravitational force between two bodies is

$$F = G \frac{M_{E,1} M_{E,2}}{r^2}.$$

Proof

This follows immediately from the relation

$$F = -m \nabla \phi.$$

12.7 Newtonian Correspondence Principle

Principle CN1

Every prediction of the PSE must reduce to a prediction of Newtonian gravitation when:

$$|h_{\mu\nu}| \ll 1, v/c \ll 1, \partial_t g_{\mu\nu} = 0.$$

Meaning

Classical gravitation constitutes the limit of minimal observable complexity of the emergent theory.

Fundamental Result

We have thus obtained the complete correspondence chain:

$$E_{\text{reg}} \rightarrow \mathcal{C} \rightarrow \Theta \rightarrow \Phi \rightarrow g_{\text{phys}} \rightarrow T_{\mu\nu} \rightarrow \text{Einstein} \rightarrow \text{Newton}.$$

General Relativity emerges as the macroscopic limit of the PSE. Newtonian gravitation emerges as the further weak-field limit of the emerging relativity.

Part VII – Emergence of Probability and Quantum Dynamics

Objective

The previous parts have shown how:

- geometry emerges from projection;
- matter emerges from projection degeneracy;
- gravitation emerges as the macroscopic limit of the emergent structure.

A fundamental question remains open: **can the emergent structure generate the probabilistic phenomena observed in quantum mechanics?**

The objective of this part is to show how probability can naturally emerge from the non-invertibility of the projection.

Chapter 13 – Emergence of Probability

13.1 Motivation

In Chapter 9, the projection fiber

$$\mathcal{F}(p) = \{E \in E_{\text{reg}} : \Phi(E) = p\}$$

was introduced. An observable configuration therefore does not identify a single emergent state. A single observation corresponds to a multiplicity of emergent structures compatible with it.

This indeterminacy constitutes the natural starting point for a theory of emerging probability.

13.2 Measure on Fibers

Definition 13.1

Let $p \in P$. Assume that on the fiber $\mathcal{F}(p)$ there is a normalized measure $\mu_{\mathcal{F}}$ such that

$$\mu_{\mathcal{F}}(\mathcal{F}(p)) = 1.$$

Interpretation

The measure represents the relative weight of the possible emergent states compatible with the same observation.

13.3 Emerging Probability

Definition 13.2

Let $A \subseteq \mathcal{F}(p)$. Define the *emerging probability*:

$$P(A) = \mu_{\mathcal{F}}(A).$$

Interpretation

Probability does not represent subjective ignorance. It measures the structural indeterminacy introduced by the projection.

Theorem 13.1 – Axioms of Probability

The function P satisfies Kolmogorov's axioms.

Proof

This follows directly from the definition of a normalized measure.

13.4 Projection Entropy and Information

Definition 13.3

Define the *informational entropy of the projection*:

$$S(P) = - \sum_i P_i \log P_i.$$

Interpretation

This quantity measures the degree of indeterminacy associated with the observation.

Theorem 13.2

The entropy is maximal when all compatible emergent states are equiprobable.

Proof

This follows from the classical theory of Shannon entropy.

13.5 Emerging Uncertainty Principle

Principle Q1

When the projection is non-invertible, the observation determines only a *distribution* of emergent states and not a unique emergent state.

Interpretation

The origin of quantum probability is traced back to the structural loss of information produced by the projection operator.

Fundamental Result

We obtain the chain:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow P \rightarrow S(P).$$

Probability therefore emerges as an effect of the degeneracy of projection fibers. It is not introduced as an independent postulate.

Chapter 14 – Emergence of the Hilbert Space and Probability Amplitudes

14.1 Motivation

In Chapter 13 it was shown that the non-invertibility of the projection naturally generates a probability measure on emergent fibers. The chain

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow P$$

was obtained.

However, probability alone is not sufficient to reproduce the structure of quantum mechanics. Quantum theory does not merely assign probabilities to events; it employs:

- probability amplitudes,
- linear superposition of states,
- interference,
- Hilbert-space structure.

The goal of this chapter is to show how these elements can emerge from the geometry of projection fibers.

14.2 Equivalent Emerging States

Recall

An observable configuration

$$p = \Phi(E)$$

does not identify a single emergent state. It determines a fiber

$$\mathcal{F}(p).$$

Interpretation

From the observational point of view, all elements of $\mathcal{F}(p)$ are indistinguishable. The available information therefore concerns not the individual emergent state, but the distribution over the entire fiber.

Definition 14.1 – Emerging Probabilistic State

The *probabilistic state* associated with the observable configuration p is the function

$$\rho: \mathcal{F}(p) \rightarrow \mathbb{R}_+$$

such that

$$\int_{\mathcal{F}(p)} \rho(E) d\mu_{\mathcal{F}} = 1.$$

Interpretation

The function ρ represents the probability distribution over the emergent states compatible with the same observation.

14.3 Square Root of Probability

Motivation

Probabilities possess an additive structure. However, the interference phenomena observed in quantum physics require a richer structure. To describe interference and superposition, a *linear representation* of probabilities must be introduced.

Definition 14.2 – Emerging Amplitude

An *emerging amplitude* is a function

$$\psi: \mathcal{F}(p) \rightarrow \mathbb{C}$$

such that

$$|\psi(E)|^2 = \rho(E).$$

Interpretation

Probability emerges as the squared modulus of the amplitude. This construction constitutes the emergent analogue of the Born rule.

Theorem 14.1 – Existence of an Amplitude Representation

For every probability distribution ρ , there exists at least one function ψ such that

$$|\psi|^2 = \rho.$$

Proof

It is sufficient to choose

$$\psi(E) = \sqrt{\rho(E)} e^{i\theta(E)},$$

where $\theta(E)$ is an arbitrary real function. Then

$$|\psi(E)|^2 = \rho(E).$$

14.4 Emerging State Space

Definition 14.3

Define the *emerging state space*:

$$\mathcal{H} = L^2(\mathcal{F}, (p)\mu_{\mathcal{F}}).$$

Interpretation

Elements of $\mathcal{H}$ are square-integrable probability amplitudes over the emergent fibers.

Theorem 14.2

The space $\mathcal{H}$ is a Hilbert space.

Proof

By construction, L^2 is complete with respect to the inner product

$$\langle \psi_1, \psi_2 \rangle = \int_{\mathcal{F}(p)} \psi_1^* \psi_2 d\mu_{\mathcal{F}}.$$

Therefore, $\mathcal{H}$ is a Hilbert space.

Corollary 14.1

The linear structure of quantum states emerges from the measurable structure of projection fibers.

14.5 Emerging Superposition Principle

Theorem 14.3

If $\psi_1, \psi_2 \in \mathcal{H}$, then

$$a\psi_1 + b\psi_2 \in \mathcal{H}$$

for all $a, b \in \mathbb{C}$.

Proof

This follows from the linearity of L^2 .

Interpretation

Quantum superposition is not postulated. It emerges directly from the linear structure of the state space.

14.6 Emerging Interference

Motivation

The most distinctive feature of quantum mechanics is interference. It must be shown how interference appears in the emergent formulation.

Theorem 14.4

Let $\psi = \psi_1 + \psi_2$. Then

$$|\psi|^2 = |\psi_1|^2 + |\psi_2|^2 + 2 \operatorname{Re}(\psi_1^* \psi_2).$$

Proof

This follows directly from the algebraic identity

$$(\psi_1 + \psi_2)^*(\psi_1 + \psi_2).$$

Interpretation

The term

$$2 \operatorname{Re}(\psi_1^* \psi_2)$$

is the interference term. Interference therefore emerges from the linear representation of distributions on fibers.

14.7 Emerging Born Rule

Theorem 14.5

The observable probability associated with a region $A \subseteq \mathcal{F}(p)$ is given by

$$P(A) = \int_A |\psi|^2 d\mu_{\mathcal{F}}.$$

Proof

This follows directly from Definitions 14.1 and 14.2.

Principle Q2

Observable probability is the squared modulus of the emerging amplitude.

Meaning

The Born rule is not introduced as an independent postulate. It emerges from the combination of:

- the structure of fibers,
- the emergent measure,
- the linear representation of states.

Chapter 15 – Emerging Quantum Dynamics

15.1 Motivation

In Chapter 14, the fundamental kinematic ingredients of the emerging quantum structure were derived:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow \rho \rightarrow \psi \rightarrow \mathcal{H}.$$

The following elements emerged:

- the Hilbert space,
- superposition,
- interference,
- the Born rule.

However, the theory is still static. A law of evolution for emerging states is missing. The purpose of this chapter is to derive a natural dynamics for emerging amplitudes and to show under which conditions it reduces to the usual Schrödinger equation.

15.2 Evolution of Emerging Fibers

Motivation

In the PSE, observable states are associated with projection fibers. Since the emergent structure evolves, the probability distribution defined on the fibers also evolves. Consequently, emerging amplitudes must depend on an evolutionary parameter.

Definition 15.1

Consider an evolutionary family of emergent states $E(\lambda)$. Define the *emerging quantum state*:

$$\psi(E, \lambda) \in \mathcal{H}.$$

Interpretation

The emerging amplitude describes the evolution of the informational distribution associated with the projection.

15.3 Conservation of Probability

Definition 15.2

Define the *total norm* of the state:

$$N(\lambda) = \int_{\mathcal{F}} |\psi(E, \lambda)|^2 d\mu_{\mathcal{F}}.$$

Interpretation

The quantity N represents the total probability available on the fiber.

Principle Q3

Physically admissible evolution must preserve total probability:

$$N(\lambda) = 1.$$

Meaning

The sum of the probabilities of the states compatible with the observation remains constant.

15.4 Evolution Generator Operator

Definition 15.3

Define an *emerging evolution generator* as a linear operator

$$\hat{H}: \mathcal{H} \rightarrow \mathcal{H}$$

that is self-adjoint.

Interpretation

The operator $\hat{H}$ measures how the hidden structure of the fiber modifies the distribution of amplitudes. In the PSE, it represents the emergent counterpart of the Hamiltonian.

Theorem 15.1 – Conservation of the Norm

Let ψ be a solution of

$$i\hbar \frac{\partial \psi}{\partial \lambda} = \hat{H}\psi, \quad \hat{H} = \hat{H}^\dagger.$$

Then $N(\lambda)$ remains constant.

Proof

Differentiating the norm:

$$\frac{dN}{d\lambda} = \int \left(\frac{\partial \psi^*}{\partial \lambda} \psi + \psi^* \frac{\partial \psi}{\partial \lambda} \right) d\mu_{\mathcal{F}}.$$

Substituting the dynamical equation and its conjugate,

$$i\hbar \frac{\partial \psi}{\partial \lambda} = \hat{H}\psi,$$

one obtains mutual cancellation of terms. Thus:

$$\frac{dN}{d\lambda} = 0.$$

15.5 Emerging Quantum Equation

Definition 15.4

Define the *fundamental emerging quantum equation*:

$$i\hbar \frac{\partial \psi}{\partial \lambda} = \hat{H}\psi.$$

Interpretation

This equation describes the evolution of the amplitude distribution associated with projection fibers. Unlike ordinary quantum mechanics:

- the state is not fundamental,
- the Hilbert space is not fundamental,
- the evolution itself derives from the structure of the projection.

15.6 Schrödinger Limit

Motivation

To verify the physical compatibility of the theory, it is necessary to show that the emerging equation can reduce to the Schrödinger equation.

Hypothesis Q1

Assume that:

- the evolutionary parameter λ coincides with observable time;
- the geometry is locally Euclidean;
- the evolution operator takes the form

$$\hat{H} = -\frac{\hbar^2}{2m}\nabla^2 + V(x).$$

Theorem 15.2 – Recovery of the Schrödinger Equation

Under Hypothesis Q1, the emerging quantum equation becomes

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi + V(x)\psi.$$

Proof

Substituting the expression for $\hat{H}$ into the emerging equation yields immediately:

$$i\hbar \frac{\partial \psi}{\partial t} = -\frac{\hbar^2}{2m}\nabla^2 \psi + V(x)\psi.$$

Corollary 15.1

Non-relativistic quantum mechanics emerges as a particular limit of the dynamics of projection fibers.

15.7 Interpretation of the Superposition Principle

Fundamental Result

Within the PSE framework, the superposition

$$a\psi_1 + b\psi_2$$

does not represent the physical coexistence of multiple observable states. It represents the superposition of different emergent configurations compatible with the same projection.

Meaning

The wave function does not directly describe fundamental reality. It describes the structural degeneracy of the projection.

15.8 Complete Quantum Chain

Collecting the results obtained in Chapters 13, 14, and 15, one obtains:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow P \rightarrow \rho \rightarrow \psi \rightarrow \mathcal{H} \rightarrow \hat{H} \rightarrow \text{Schrodinger}.$$

Part VIII – Emerging Unification of Gravity and Quantum Theory

Objective

The previous parts have shown that two fundamental structures of physics can emerge from the same architecture of the PSE.

On one side:

$$E_{\text{reg}} \rightarrow \mathcal{C} \rightarrow \Theta \rightarrow \Phi \rightarrow g_{\text{phys}} \rightarrow T_{\mu\nu} \rightarrow \text{Einstein} \rightarrow \text{Newton}.$$

On the other side:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow \mathcal{F}(p) \rightarrow P \rightarrow \rho \rightarrow \psi \rightarrow \mathcal{H} \rightarrow \hat{H} \rightarrow \text{Schrodinger}.$$

At first sight, the two constructions appear independent. The purpose of this part is to show that both derive from the same fundamental structure:

$$E_{\text{reg}}.$$

Chapter 16 – Principle of Emerging Unification

16.1 Motivation

In traditional physical theories:

- gravity is described by the geometry of space-time;
- quantum mechanics is described by states in Hilbert space.

The two structures possess deeply different mathematical formulations. In the PSE, however, both emerge from the same fundamental operation:

$$\Phi: E_{\text{reg}} \rightarrow \mathcal{P}.$$

This observation suggests the existence of a deeper unifying principle.

16.2 Dual Manifestation of the Projection

Definition 16.1

Define:

- **Geometric Sector**

$$\mathbb{G} = \{g_{\text{phys}}, T_{\mu\nu}\};$$

- **Quantum Sector**

$$\mathbb{Q} = \{\mathcal{H}, \psi, \hat{H}\}.$$

Interpretation

The geometric sector describes what *survives* the projection. The quantum sector describes the information *hidden* in the projection fiber.

Fundamental Observation

Both sectors derive from the same operator:

$$\Phi.$$

Indeed:

$$\Phi \Rightarrow g_{phys}, \Phi \Rightarrow \mathcal{F}(p) \Rightarrow \mathcal{H}.$$

16.3 Principle of Emerging Duality

Principle U1

Observable geometry and quantum structure represent two complementary aspects of the same emergent projection.

Interpretation

Within the PSE:

- geometry describes the information *preserved* by the projection;
- quantum theory describes the information *lost* through the projection.

Conceptual Scheme

$$E_{\text{reg}} \xrightarrow{\Phi} \mathcal{P} \text{ produces simultaneously } g_{phys} \text{ and } \mathcal{H}.$$

16.4 Total Amount of Information

Definition 16.2

Define the *total structural content*:

$$\mathcal{I}(E) = I_{geom}(E) + I_{quant}(E),$$

where:

- I_{geom} measures the information preserved by the projection,
- I_{quant} measures the information hidden in the fibers.

Interpretation

The projection does not destroy information. It redistributes it between:

- the observable sector,
- the hidden sector.

Theorem 16.1 – Structural Conservation

Let E be an emergent state. Then $\mathcal{I}(E)$ remains invariant under projection.

Proof

By construction:

- the preserved information generates the geometric sector;
- the non-preserved information generates the fibers.

The sum of the two components coincides with the original structural content. Therefore:

$$\mathcal{I}(E) = \text{constant.}$$

16.5 Common Origin of Mass and Probability

Theorem 16.2

Emerging mass and emerging probability both derive from projection degeneracy.

Proof

From Chapter 9:

$$\mathcal{F}(p) \rightarrow \rho_{\Phi} \rightarrow T_{\mu\nu}.$$

From Chapter 13:

$$\mathcal{F}(p) \rightarrow P.$$

Both constructions depend on the same fiber. They therefore possess a common structural origin.

Interpretation

In the PSE:

- mass emerges from the *density* of degeneracy;
- probability emerges from the *measure* of degeneracy.

They are thus different manifestations of the same phenomenon.

16.6 Principle of Emerging Unification

Principle U2

Gravity and quantum dynamics emerge from the same structural emergent space and from the same class of projection operators.

Symbolic Form

$$E_{\text{reg}} \Rightarrow \Phi \Rightarrow \begin{cases} \mathcal{G}_{phys} \rightarrow T_{\mu\nu} \\ \mathcal{H} \rightarrow \psi \end{cases}$$

Fundamental Result

The distinction between:

- geometry and
- quantum theory

is **not fundamental**. It appears only *after* projection. The primary structure is instead:

$$E_{\text{reg}}.$$

Chapter 17 – Critical Regimes and Transitions Between Geometry and Quantum Theory

17.1 Motivation

In Chapter 16 it was shown that gravity and quantum dynamics both emerge from the same fundamental structure

$$E_{\text{reg}}.$$

However, the two sectors appear very different. In the macroscopic regime, one observes:

$$g_{\text{phys}}, T_{\mu\nu},$$

while in regimes dominated by fiber degeneracy one observes:

$$\mathcal{H}, \psi, \hat{H}.$$

The purpose of this chapter is to identify the parameters that determine which of the two sectors becomes dominant.

17.2 Degeneracy Ratio

Definition 17.1

Let $E \in E_{\text{reg}}$. Define the *emergent degeneracy index*:

$$\Delta(E) = \frac{I_{\text{quant}}(E)}{I_{\text{geom}}(E)}.$$

Interpretation

The quantity Δ measures the ratio between:

- information hidden in the fibers,
- information preserved by the projection.

Meaning

Small values of Δ indicate geometric dominance. Large values of Δ indicate quantum dominance.

17.3 Geometric Regime

Definition 17.2

A system is said to be in the *geometric regime* when

$$\Delta(E) \ll 1.$$

Interpretation

Most of the information survives the projection. The fibers possess reduced degeneracy.

Theorem 17.1

In the geometric regime, the observable dynamics is dominated by the gravitational sector.

Proof

If $\Delta(E) \ll 1$, then $I_{quant} \ll I_{geom}$. The observable structure is therefore determined predominantly by geometric invariants. The contribution of the fibers is negligible. Hence the emergent equations reduce to the geometric sector.

Corollary 17.1

In the limit $\Delta(E) \rightarrow 0$, General Relativity is recovered.

17.4 Quantum Regime

Definition 17.3

A system is said to be in the *quantum regime* when

$$\Delta(E) \gg 1.$$

Interpretation

The predominant part of the emergent structure is hidden by the projection. The fibers become dominant.

Theorem 17.2

In the quantum regime, the observable dynamics is dominated by the emerging Hilbert space.

Proof

If $I_{quant} \gg I_{geom}$, the most efficient description of the system is given by distributions on the fibers. The emerging amplitudes ψ therefore provide the natural language of the theory.

Corollary 17.2

In the limit $\Delta(E) \rightarrow \infty$, Schrödinger dynamics is recovered.

17.5 Critical Regime

Motivation

The most interesting situation occurs when neither sector dominates completely.

Definition 17.4

The *critical unification regime* is the region of emergent space in which

$$\Delta(E) \simeq 1.$$

Interpretation

In this regime:

$$I_{geom} \sim I_{quant}.$$

Observable geometry and fiber structure contribute comparably.

Principle U3

Unification between gravity and quantum theory manifests in critical regimes characterized by

$$\Delta(E) \simeq 1.$$

Meaning

Neither the geometric description nor the quantum description is sufficient on its own. Both must be used simultaneously.

17.6 Emerging Critical Length

Definition 17.5

Define the *emerging critical scale* ℓ_c as the characteristic length at which

$$\Delta(E) = 1.$$

Interpretation

The quantity ℓ_c separates:

- the macroscopic geometric regime,
- the microscopic quantum regime.

Theorem 17.3

For scales larger than ℓ_c , the gravitational sector dominates. For scales smaller than ℓ_c , the quantum sector dominates.

Proof

This follows directly from the definition of $\Delta(E)$ and the definition of the critical length.

17.7 Diagram of Emerging Regimes

The overall structure of the theory can be represented by the following scheme:

$$\Delta \ll 1 \rightarrow \Delta \simeq 1 \rightarrow \Delta \gg 1$$

$$\text{Gravity} \rightarrow \text{Unification} \rightarrow \text{Quantum}$$

Fundamental Result

The distinction between:

- space-time and
- quantum state

is **not absolute**. It depends on the ratio between preserved geometric information and hidden fiber information.

Chapter 18 – Origin of the Emerging Critical Scale

18.1 Motivation

In Chapter 17 the parameter

$$\Delta(E) = \frac{I_{quant}(E)}{I_{geom}(E)}$$

and the corresponding critical length

$$\ell_c, \Delta(E) = 1,$$

were introduced. This quantity separates:

- the gravitational regime,
- the quantum regime,
- the unification regime.

However, the scale ℓ_c was introduced only qualitatively. For the PSE to constitute a complete theory, it is necessary to show how this scale emerges from the internal structure of the theory.

18.2 Geometric Complexity and Quantum Complexity

Definition 18.1

Define:

- **Geometric Complexity**

$$C_G(E) = I_{geom}(E),$$

- **Quantum Complexity**

$$C_Q(E) = I_{quant}(E).$$

Interpretation

These two quantities measure respectively:

- the organization preserved by the projection,
- the organization hidden in the fibers.

Observation

By Principle U1:

$$C_G + C_Q = C_{tot}.$$

The total information of the emergent structure is simply redistributed.

18.3 Informational Equilibrium Point

Definition 18.2

An *informational equilibrium point* is any state E_c such that

$$C_G(E_c) = C_Q(E_c).$$

Interpretation

In such a configuration, preserved and hidden information are exactly balanced.

Theorem 18.1

Every informational equilibrium point satisfies

$$\Delta(E_c) = 1.$$

Proof

By definition:

$$\Delta(E_c) = \frac{C_Q(E_c)}{C_G(E_c)}.$$

Since

$$C_Q(E_c) = C_G(E_c),$$

it follows immediately that

$$\Delta(E_c) = 1.$$

18.4 Origin of the Critical Scale

Definition 18.3

Define the *emerging critical scale* as the characteristic length associated with the configuration E_c . It is denoted by

$$\ell_c.$$

Interpretation

The critical length is not introduced as a fundamental constant. It emerges from the balance between geometric structure and quantum structure.

Principle U4

The emerging critical scale of the theory coincides with the scale at which geometric complexity and quantum complexity become equal.

Symbolic Form

$$\ell_c \Leftrightarrow C_G = C_Q.$$

18.5 Stability of the Critical Scale

Motivation

A physically meaningful scale cannot depend on arbitrary perturbations.

Theorem 18.2

If the functionals C_G and C_Q are continuous, then the critical scale is stable under small perturbations of the emergent state.

Proof

The quantities $C_G(E)$ and $C_Q(E)$ vary continuously. Therefore, the point defined by

$$C_G(E) = C_Q(E)$$

moves continuously as E varies.

Hence, the associated critical length does not exhibit spontaneous discontinuities.

Corollary 18.1

The transition between the geometric sector and the quantum sector occurs smoothly.

18.6 Interpretation of the Planck Scale

Fundamental Observation

In conventional physics, the Planck length

$$\ell_P = \sqrt{\frac{\hbar G}{c^3}}$$

is constructed by combining fundamental constants. In the PSE, the primary quantity is instead the ratio

$$\Delta(E).$$

Correspondence Hypothesis

If the PSE correctly reproduces known physics, then the emerging critical length must satisfy

$$\ell_c \sim \ell_P$$

in the observable macroscopic limit.

Interpretation

The Planck length therefore appears as the physical manifestation of the emerging critical scale, and not as an independent fundamental constant.

18.7 Unification Region

Definition 18.4

Define the *unification region*:

$$\mathfrak{U} = \{E: \Delta(E) \simeq 1\}.$$

Interpretation

In this region:

$$C_G \sim C_Q.$$

Neither a purely gravitational nor a purely quantum description is sufficient.

Theorem 18.3

In the region $\mathfrak{U}$, the two emergent sectors must be treated simultaneously.

Proof

This follows directly from the definition $\Delta(E) \simeq 1$. In this regime, the two informational components have equal structural importance.

Fundamental Result

The critical scale is not postulated. It emerges from the theory through the condition

$$C_G(E) = C_Q(E).$$

Thus one obtains the chain:

$$E_{\text{reg}} \rightarrow \Phi \rightarrow (C_G, C_Q) \rightarrow \Delta(E) \rightarrow \ell_c.$$

Chapter 19 – Dynamics of Unification and Predictions of the PSE

19.1 Motivation

In Chapter 18, the emerging critical scale

$$\ell_c, C_G(E) = C_Q(E),$$

was derived. This scale separates the sector dominated by geometry from the sector dominated by quantum structure.

A fundamental question remains: **how does a system dynamically evolve when it crosses the unification region?**

The purpose of this chapter is to construct a dynamics of unification and to identify the first possible observable physical consequences of the PSE.

19.2 Unification Parameter

Definition 19.1

Introduce the *unification parameter*:

$$U(E) = \frac{C_Q(E) - C_G(E)}{C_Q(E) + C_G(E)}.$$

Properties

By construction:

$$-1 \leq U(E) \leq 1.$$

Moreover:

- $U(E) \approx -1$: purely geometric regime,
- $U(E) \approx 0$: critical regime,
- $U(E) \approx 1$: quantum regime.

Interpretation

The quantity $U(E)$ measures the position of the system along the gravity–quantum axis.

19.3 Dynamics of Unification

Definition 19.2

Consider an evolutionary family $E(\lambda)$. Define the *unification trajectory*:

$$U(\lambda) = U(E(\lambda)).$$

Interpretation

The evolution of the emergent structure induces motion between the different regimes of the theory.

Principle U5

Emergent evolution tends to reduce extreme imbalances between geometric complexity and quantum complexity.

Mathematical Form

Introduce a *unification potential* $W(U)$. Trajectories satisfy:

$$\frac{d^2 U}{d\lambda^2} + W'(U) = 0.$$

Meaning

The theory naturally tends toward regions in which the overall structure is more balanced.

19.4 Stability of the Critical Region

Definition 19.3

A point U_c is an *unification equilibrium* if

$$W'(U_c) = 0.$$

Theorem 19.1

If

$$W''(U_c) > 0,$$

then the critical regime is stable.

Proof

Linearizing:

$$U = U_c + \varepsilon,$$

one obtains:

$$\ddot{\varepsilon} + W''(U_c) \varepsilon = 0.$$

Solutions remain bounded. Thus the configuration is stable.

Corollary 19.1

It is possible for physical structures to be naturally localized near the unification region.

19.5 Emerging Corrections to Gravity

Motivation

In ordinary regimes $\Delta(E) \ll 1$, General Relativity is recovered. However, near the critical scale additional contributions may appear.

Definition 19.4

Introduce the *correction term* $\Xi(U)$, depending on the degree of unification. The geometric equations take the general form:

$$R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu} = \kappa T_{\mu\nu} + \Xi(U).$$

Interpretation

When $U \simeq 0$, the quantum structure of the fibers directly influences geometry.

Corollary 19.2

In the limit $U \rightarrow -1$, the Einstein equations are exactly recovered.

19.6 Emerging Corrections to Quantum Dynamics

Motivation

The same mechanism must apply in the quantum sector. Residual geometric information may modify Schrödinger dynamics.

Definition 19.5

Introduce a *correction operator* $\hat{\Gamma}(U)$. The general quantum equation becomes:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H}\psi + \hat{\Gamma}(U)\psi.$$

Interpretation

Deviations from ordinary quantum mechanics depend on the system's proximity to the unification regime.

Corollary 19.3

In the limit $U \rightarrow 1$, the standard Schrödinger equation is recovered.

19.7 First Qualitative Predictions of the PSE

Fundamental Result

The theory suggests the existence of a physical region in which:

- geometry is not fully classical,
- quantum dynamics is not fully standard,
- the two structures simultaneously influence observable evolution.

Possible Emerging Phenomena

In principle, the PSE admits:

- gravitational corrections to quantum dynamics,
- quantum corrections to geometry,
- deviations from Einstein's equations near ℓ_c ,
- deviations from Schrödinger's equation near ℓ_c ,
- new structural states localized in the critical region.

19.8 Diagram of the Complete Architecture

Collecting the entire theory:

$$E_{\text{reg}} \rightarrow C \rightarrow \Theta \rightarrow \Phi \rightarrow (g_{\text{phys}}, T_{\mu\nu}, \mathcal{H}, \psi) \rightarrow \Delta \rightarrow \ell_c \rightarrow U,$$

and therefore:

$$U \Rightarrow \begin{cases} \text{Einstein} + \Xi(U), \\ \text{Schrodinger} + \Gamma(U). \end{cases}$$

Chapter 20 – General Conclusions, Limits of the Theory, and Verification Program

20.1 Motivation

This monograph has developed an advanced mathematical formulation of the PSE with the goal of constructing an emergent theory capable of describing, within a single formal architecture:

- complexity,
- projection,
- geometry,
- matter,
- probability,
- quantum dynamics,
- unification.

This chapter summarizes the results obtained, identifies the current limits of the theory, and defines the main future development directions.

20.2 Fundamental Results

First Result

A fundamental emergent space was constructed:

$$E_{\text{reg}} = C^0(X \times X) \times C^0(X \times X),$$

on which the quantities D_e^{reg} , C_e^{reg} are defined. All subsequent structures derive from this space.

Second Result

A global complexity function was constructed:

$$C(E) = \alpha_D C_D + \alpha_C C_C + \alpha_T C_T.$$

This function is:

- well-defined,
- finite,
- continuous,
- structurally invariant.

Third Result

The thresholds $\Theta_1, \Theta_2, \dots, \Theta_n$ were not introduced as primitive constants. They emerge as critical values of complexity:

$$\Theta_k = \mathcal{C}(E_{\lambda_k}).$$

Fourth Result

Projection operators were constructed:

$$\Phi: E_{\text{reg}} \rightarrow \mathcal{P}.$$

Their fundamental properties are:

- continuity,
- monotonicity,
- compressivity,
- non-invertibility.

Fifth Result

From the invariants of the projection, a geometry emerges:

$$(\mathcal{M}_E, g).$$

Through the macroscopic limit:

$$g \rightarrow g_{\text{phys}},$$

a physical emergent metric is obtained.

Sixth Result

The degeneracy of projection fibers generates:

$$\mathcal{F}(p) \rightarrow \rho_\Phi \rightarrow \Psi \rightarrow T_{\mu\nu}.$$

In the PSE, matter therefore emerges from hidden informational structure.

Seventh Result

In the macroscopic regime, the equations

$$R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu} = \kappa T_{\mu\nu}$$

are recovered. General Relativity thus appears as an effective theory of the emergent structure.

Eighth Result

In the weak-field limit, one obtains:

$$\nabla^2 \phi = 4\pi G \rho_E.$$

Newtonian gravitation is recovered as a further limit.

Ninth Result

The non-invertibility of the projection naturally produces:

$$P \rightarrow \rho \rightarrow \psi \rightarrow \mathcal{H}.$$

The following were derived:

- emerging probability,
- emerging Hilbert space,
- emerging Born rule,
- emerging superposition.

Tenth Result

Fiber dynamics leads to the equation:

$$i\hbar \frac{\partial \psi}{\partial t} = \hat{H} \psi.$$

The Schrödinger equation is obtained as a particular limit.

Eleventh Result

Gravity and quantum theory derive from a common structure:

$$E_{\text{reg}}.$$

The distinction between geometric and quantum sectors emerges only *after* projection.

20.3 Final Architecture of the Theory

The complete structure of the PSE can be summarized by the chain:

$$E_{\text{reg}} \rightarrow C \rightarrow \Theta \rightarrow \Phi \rightarrow (g_{\text{phys}}, T_{\mu\nu}, \mathcal{H} \psi) \rightarrow (\text{Einstein, Schrodinger})$$

and subsequently:

$$(g_{\text{phys}}, T_{\mu\nu}, \mathcal{H} \psi) \rightarrow \Delta \rightarrow \ell_c \rightarrow U.$$

20.4 Current Limits of the Theory

Despite the results obtained, the PSE still presents several aspects requiring development.

Limit L1

The theory is formulated in very general terms. A fully computable explicit model of the emergent space is still missing.

Limit L2

The functions $D_{S,\text{reg}}^{(e)}$ and $C_{e,\text{reg}}$ have been treated axiomatically. Concrete examples must be identified.

Limit L3

The correction terms $\Xi(U)$, $\Gamma(U)$ were introduced in general form. Their explicit structure remains to be determined.

Limit L4

The theory does not yet provide unique quantitative predictions directly comparable with experiments.

20.5 Program for Physical Verification

For the PSE to become a physically testable theory, at least four research programs must be developed.

Program P1 – Construction of Explicit Models

Concrete examples of X , $D_{S,\text{reg}}^{(e)}$, $C_{e,\text{reg}}$ must be constructed.

Program P2 – Calculation of the Critical Scale

The quantity

$$\ell_c$$

must be determined quantitatively, and it must be verified whether

$$\ell_c \approx \ell_P.$$

Program P3 – Corrections to Einstein's Equations

The term $\Xi(U)$ must be explicitly determined, and observable deviations from General Relativity must be sought.

Program P4 – Corrections to the Schrödinger Equation

The operator $\Gamma(U)$ must be determined, and possible experimental effects must be identified.

20.6 Final Principle of the PSE

All results of the theory can be summarized in the following principle.

PSE Principle

Observable physical reality is not the fundamental level of structure. It emerges through projection operators that transform a primary emergent structure

$$E_{\text{reg}}$$

into:

- observable geometry,
- observable matter,
- observable probability,
- observable dynamics.

20.7 General Conclusion

The advanced mathematical formulation of the PSE leads to a unified vision in which:

$$\boxed{\text{Geometry} = \text{preserved information}}$$

and

$$\boxed{\text{Quantum theory} = \text{hidden information}}$$

are two complementary manifestations of the same emergent structure.

Within the PSE:

- space-time is not fundamental,
- matter is not fundamental,
- probability is not fundamental,
- the wave function is not fundamental.

The primary emergent regularized structure is:

$$\boxed{E_{\text{reg}}}.$$

From it, through complexity and projection, all observable physical objects emerge.